INTACT STABILITY CALCULATION

CGS RUPOSHI BANGLA

Contents

General Information

Principal Particulars:

Length Overall	: 38.50 M
Breadth molded	: 7.00 M
Depth at mid	: 4.08 M
Designed draft	: 1.75 M
Displacement	: 195.00 T

Effect of New Engine Installation:

The old engine weighed 5,161 kg, whereas the new engine weighs 7,910 kg—an increase of 2.749 tons per engine, totaling 5.498 tons of added weight to the ship. This additional weight was included in the stability analysis to evaluate its impact on the vessel's overall stability and equilibrium.

USE OF THIS BOOKLET

This stability booklet is an integral part of the ship's documentation and must always be kept on board. It should be complete, clearly legible, and easily accessible for reference. In the event that the booklet is lost or damaged, a replacement must be acquired without delay. The information within remains applicable unless it is officially replaced by an updated booklet or directive.

The loading scenarios included here reflect common operational conditions. However, for any loading condition that differs from those shown, a separate stability calculation is required. The Master must always adhere to the ship's established operational procedures.

OPERATING RESTRICTIONS

The hydrostatic data provided in this booklet is based on the vessel's design trim. Significant deviations from this trim can cause actual hydrostatic values to differ from those listed, potentially leading to an overestimation of the vessel's stability. Therefore, maintaining the vessel as close as possible to its design trim is essential.

Additionally, the total volume of liquids in the tanks must not fall below the specified MINOP condition. The MINOP loading state refers to the following tank configuration: Tank 1 at 50%, Tank 2 at X%... representing approximately 30% of the total capacity for each liquid group.

MASTER'S SHIPBOARD PROCEDURES

It is the Commanding Officer's duty to ensure the vessel's watertight integrity, which is crucial for maintaining optimal stability and reserve buoyancy in the event of underwater damage. All watertight doors and hatches must be secured while at sea and closed promptly after use. In cases where heavy following seas or adverse weather are expected, the vessel should either head into the seas or navigate to sheltered waters for safety.

SPECIAL NOTES

Tank Usage and Free Surface Effects:

When a tank is completely filled with liquid, there is no movement within it, and its impact on ship stability is equivalent to that of solid cargo. However, once any amount of liquid is removed, the condition changes significantly due to the *free surface effect*, which negatively affects the vessel's stability. This phenomenon results in a loss of GM (metacentric height) or a *virtual rise in the vessel's vertical center of gravity (VCG)*, and is calculated as:

Loss of GM = Free Surface Moment (tonne·m) / Vessel Displacement (tonnes):

While preparing loading conditions, the free surface effect must be considered for all tanks that are either partially filled (slack) or are expected to become slack. At departure, this typically includes all main fuel and fresh water tanks. To minimize the adverse impact on stability:

- Keep the number of slack tanks as low as possible.
- Close cross-connections between port and starboard tanks while at sea.
- Ensure ballast tanks are either fully filled (*pressed up*) or completely empty.
- Minimize dirty water accumulation in bilges.

Use of KN Curves in Gusting Conditions:

When the vessel is under a steady wind, it heels to the angle where the wind's heeling arm curve intersects the GZ (righting arm) curve. However, if a gust strikes, the heel angle increases to a new intersection point between the gust-induced heeling arm curve and the GZ curve. Note that the heeling moment increases with the *square* of the apparent wind speed.

Wind Heeling Arm Calculations:

Additional details should be inserted here, depending on how the wind heeling arms are to be calculated

$$l_{w_1} = \frac{P * A * Z}{1000 * g * \Delta} \quad (m) \quad \text{and}$$

$$l_{w_2} = 1.5 * l_{w_1} \quad (m)$$

where:

P = wind pressure of 504 Pa. The value of P used for ships in restricted service may be reduced subject to the approval of the Administration

A = projected lateral area of the portion of the ship and deck cargo above the waterline (m^2)

Z = vertical distance from the centre of A to the centre of the underwater lateral area or approximately to a point at one half the mean draught (m)

Δ = displacement (t)

g = gravitational acceleration of 9.81 m/s^2.

OPERATIONAL PROVISIONS AGAINST CAPSIZING

(According to IS Code 2008, Part B, Chapter 5)

General Precautions to Prevent Capsizing:

- Meeting the minimum stability requirements does not guarantee immunity from capsizing. The Master must always act with caution and sound judgment, taking into account seasonal weather, forecasts, and the navigational area. Adjustments to speed and course should be made based on prevailing conditions.
- Cargo should be planned and stowed in a way that maintains compliance with stability criteria. If necessary, limit cargo volume to allow for additional ballast weight.
- Prior to departure, ensure that all cargo, cranes, and large equipment are properly secured or lashed to prevent movement due to rolling and pitching during the voyage.
- Minimize the number of slack (partially filled) tanks, as they adversely impact stability. Also, consider the negative effects of pool tanks on stability.
- Although the criteria set minimum GM (metacentric height) values, high GM values can also be problematic due to excessive acceleration forces, which may harm the ship, crew, cargo, or equipment. Slack tanks may be used to reduce GM, but the resulting sloshing effects must be considered carefully.
- Special attention is required when carrying certain bulk cargoes, as some may have adverse effects on stability. Refer to the IMO *Code of Safe Practice for Solid Bulk Cargoes* for guidance.

Precautions in Heavy Weather:

- Ensure all openings (doors, hatches, vents) that could allow water ingress into the hull or superstructure are securely closed during rough weather. Equipment for securing these should always be in working order and readily available.
- Watertight and weathertight closures should remain shut during navigation, except when necessary for ship operations. They should be ready for immediate closure and clearly marked. Hatch covers and deck scuttles, especially on fishing vessels, must be properly fastened when not in use. Deadlights should also be kept secure and functional in adverse conditions.
- Closing devices for fuel tank vent pipes must be secured in heavy weather.

Ship Handling in Heavy Weather:

- Maintain sufficient freeboard under all loading conditions to ensure seaworthiness.
- In rough seas, reduce speed if the vessel experiences excessive propeller emergence, deck flooding, or hull slamming.
- Exercise extreme caution when navigating in following, quartering, or head seas. Hazardous phenomena such as **parametric rolling, broaching-to, reduced crest stability**, and **excessive rolling** can occur singly or in combination, posing a risk of capsizing. Adjust speed and/or course to avoid these dangers.
- Avoid overreliance on automatic steering, as it may hinder rapid course changes required in bad weather.
- Prevent water from collecting in deck wells. If freeing ports are inadequate for drainage, reduce speed, change course, or both. Closing devices on freeing ports must always function properly and should never be locked shut.
- Be aware of the risk of steep or breaking waves in specific regions, such as estuaries, shallow waters, or funnel-shaped bays—these conditions can be especially hazardous for smaller vessels.
- Strong lateral wind pressure can cause significant heeling. While anti-heeling measures may correct this, improper use or changes in wind direction can lead to unsafe heel angles or capsizing. Such measures should only be used if the vessel has been approved—by calculation and administration—for safe operation under worst-case conditions. Usage guidelines must be included in the stability booklet.
- Operational guidance—such as onboard decision-support systems or weather routing software—should be provided to help avoid dangerous situations. These tools should be simple and user-friendly.

UNIT CONVERSIONS

Multiply by	To convert from	To obtain	
0.039370	mm	inches	25.4
0.39370	cm	inches	2.54
3.2808	m	feet	0.3048
2.2046	kg	lbs	0.45359
0.00098421	kg	Tons (2240 lbs)	1016.0
0.98421	Tonnes (1000 KG)	Tons (2240 lbs)	1.016
2.4999	Tonnes per cm	Tons per inch	0.40002
8.2017	Tonnes metres units (MCTC)	Ton feet units (MCTI)	0.12193
187.98	Metre Radians	Foot Degree	0.0053198

RELATIONSHIPS BETWEEN WEIGHT AND VOLUME

10mm cubed = 1 cubic cm
1 cubic cm of fresh water (S.G. = 1.0) = 1 gram
1000 cubic cm of fresh water (S.G. 1 = 1.0) = 1 kg (1000grams)
1 cubic meter of fresh water (S.G. 1 = 1.0) = 1 tonnes (1000kg)
1 cubic meter of salt water (S.G. 1 = 1.025) = 1.025 tonnes (1025kg)
1 Tonne of salt water (S.G. 1 = 1.025) = 0.975 cubic metres
1 cubic metre = 35.316 cubic feet
1 cubic foot = 0.0283 cubic metres

FRAME OF REFERENCE

X: Aft Perpendicular (forward +ve)

Y: Centre line (starboard +ve)

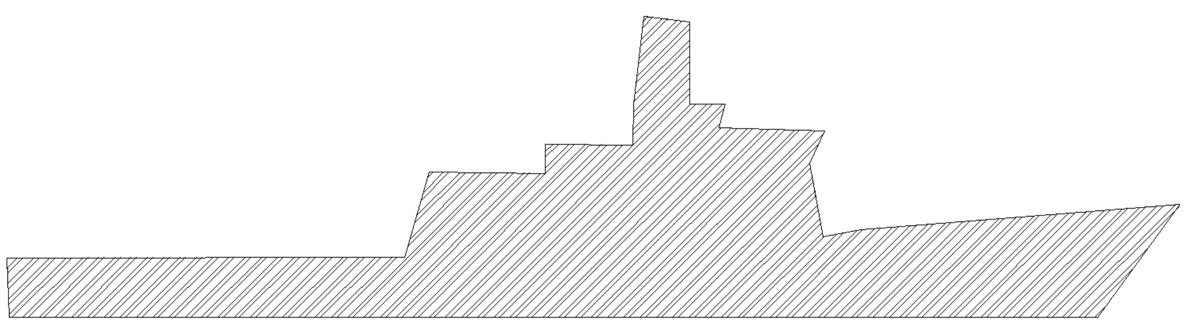
Z: Base Line (upward +ve)

COMPLIANCE REQUIREMENT

As this ship is required to comply with IMO requirements for statical and dynamical stability, it is most important to ensure that, in any sailing condition, the stability complies at least with the following minimum criteria:

- 2.2.1 The area under the righting lever curve (GZ curve) shall not be less than 0.055 meter-radians up to φ = 30° angle of heel and not less than 0.09 meter-radians up to φ = 40° or the angle of down-flooding φ_f footnote if this angle is less than 40°. Additionally, the area under the righting lever curve (GZ curve) between the angles of heel of 30° and 40° or between 30° and φ_f, if this angle is less than 40°, shall not be less than 0.03 meter-radians.
- 2.2.2 The righting lever GZ shall be at least 0.2 m at an angle of heel equal to or greater than 30°.
- 2.2.3 The maximum righting lever shall occur at an angle of heel not less than 25°. If this is not practicable, alternative criteria, based on an equivalent level of safety, may be applied subject to the approval of the Administration.
- 2.2.4 The initial metacentric height GM_0 shall not be less than 0.15
- 2.3 IMO weather criteria

WINDAGE AREA



Total Area: 138.87 M^2

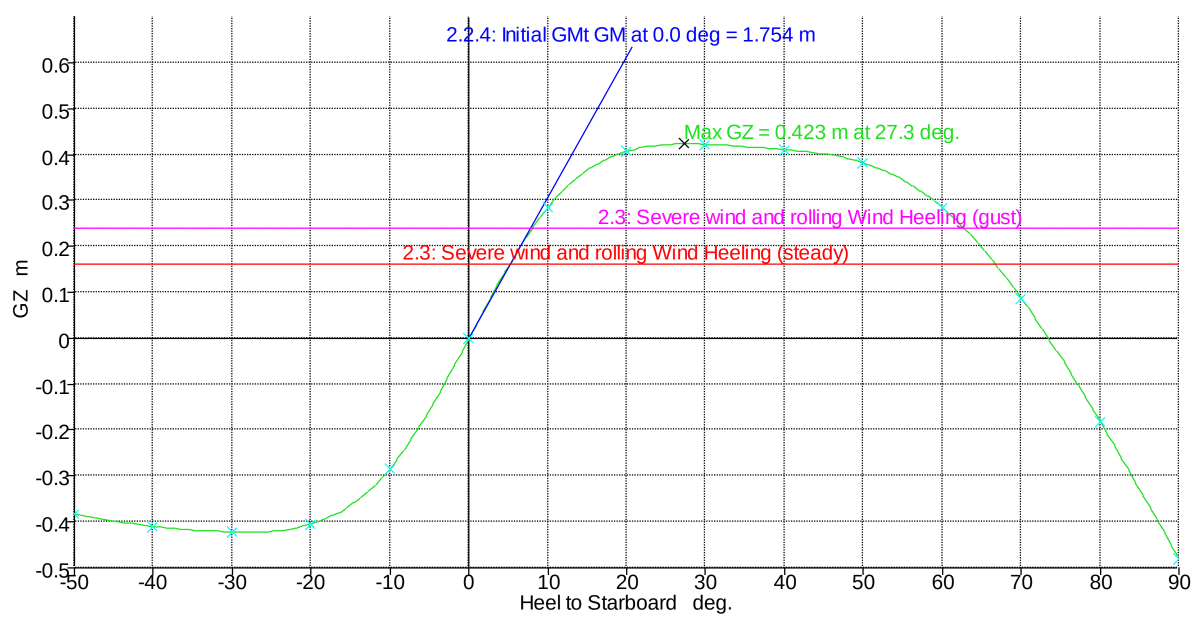
Area centroid from baseline:4.26 M

Stability Calculation

Lightship

Item Name	Quantity	Unit Mass tonne	Total Mass tonne	Long. Arm m	Vert. Arm m	Total FSM tonne.m
Lightship	1	131.830	131.830	16.540	3.360	0.000
Crew & Effects	1	4.000	4.000	20.000	4.100	0.000
Provisions	1	1.000	1.000	16.000	2.500	0.000
Ammunition	1	1.500	1.500	3.500	2.500	0.000
Liquid in systems	1	3.500	3.500	12.000	1.500	0.000
Additional Weight of new Engine	1	5.498	5.498	12.200	1.240	0.000
Total Loadcase			147.328	16.228	3.242	0.000
FS correction					0.000	
VCG fluid					3.242	

Draft Amidships m	1.589
Displacement t	147.3
Heel deg	0.0
Draft at FP m	1.483
Draft at AP m	1.695
Trim (+ve by stern) m	0.211
LCB from zero pt. (+ve fwd) m	16.214
LCF from zero pt. (+ve fwd) m	14.722
KB m	1.126
GMt corrected m	1.754
KML m	106.790
Immersion (TPc) tonne/cm	1.887
MTc tonne.m	4.252



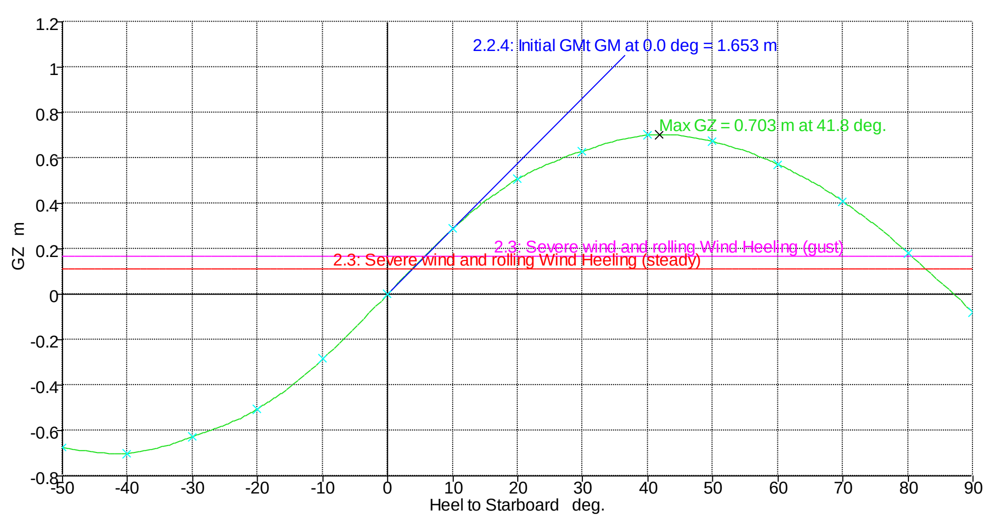
Heel to Starboard deg	0.0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	80.0	90.0
GZ m	0.001	0.285	0.406	0.422	0.410	0.383	0.285	0.087	-0.182	-0.480
Area under GZ curve from zero heel m.deg	0.0015	1.5220	5.1119	9.3028	13.4722	17.4681	20.8863	22.8262	22.3933	19.0926
Displacement t	147.3	147.3	147.3	147.3	147.3	147.3	147.3	147.3	147.3	147.3
Draft at FP m	1.484	1.456	1.413	1.325	1.139	0.792	0.170	-1.222	-5.618	n/a
Draft at AP m	1.694	1.673	1.513	1.207	0.735	0.035	-1.008	-2.879	-8.142	n/a
WL Length m	35.639	35.617	35.581	35.504	35.326	34.900	34.238	35.588	36.319	36.879
Beam max extents on WL m	6.830	6.374	5.837	5.712	5.721	5.747	5.643	4.397	3.768	3.507
Wetted Area m^2	221.649	216.573	206.009	202.030	202.761	207.006	208.859	205.124	203.788	203.637
Waterpl. Area m^2	188.671	181.429	166.826	161.337	163.094	155.911	143.270	125.547	114.862	108.800
Prismatic coeff. (Cp)	0.669	0.667	0.668	0.673	0.680	0.702	0.744	0.752	0.777	0.797
Block coeff. (Cb)	0.385	0.426	0.475	0.461	0.430	0.425	0.432	0.530	0.602	0.614
LCB from zero pt. (+ve fwd) m	16.215	16.215	16.223	16.234	16.248	16.262	16.269	16.268	16.259	16.245
LCF from zero pt. (+ve fwd) m	14.722	15.114	16.047	16.752	17.363	18.092	18.706	18.375	18.134	17.938
Max deck inclination deg	0.3367	10.0058	20.0005	30.0004	40.0025	50.0044	60.0045	70.0026	80.0008	90.0000
Trim angle (+ve by stern) deg	0.3367	0.3464	0.1592	-0.1883	-0.6443	-1.2087	-1.8803	-2.6440	-4.0232	n/a

Code	Criteria	Value	Units	Actual	Status	Margin %
267(85) Ch2 - General Criteria	2.3: IMO roll back angle	16.8	deg			
267(85) Ch2 - General Criteria	2.2.1: Area 0 to 30	3.1513	m.deg	9.3028	Pass	+195.20
267(85) Ch2 - General Criteria	2.2.1: Area 0 to 40	5.1566	m.deg	13.4722	Pass	+161.26
267(85) Ch2 - General Criteria	2.2.1: Area 30 to 40	1.7189	m.deg	4.1694	Pass	+142.56
267(85) Ch2 - General Criteria	2.2.2: Max GZ at 30 or greater	0.200	m	0.422	Pass	+111.00
267(85) Ch2 - General Criteria	2.2.3: Angle of maximum GZ	25.0	deg	27.3	Pass	+9.09
267(85) Ch2 - General Criteria	2.2.4: Initial GMt	0.150	m	1.754	Pass	+1069.33
267(85) Ch2 - General Criteria	2.3: Severe wind and rolling				Pass	
	Angle of steady heel shall not be greater than (<=)	16.0	deg	5.1	Pass	+67.93
	Angle of steady heel / Deck edge immersion angle shall not be greater than (<=)	80.00	%	12.48	Pass	+84.40
	Area1 / Area2 shall not be less than (>=)	100.00	%	109.98	Pass	+9.98
	Area 1 shall not be less than (>=)	0.0000	m.deg	6.3533	Pass	infinite
Regulation 28 GZ-based	28.3.2 Equi heel <= 25 or <= 30 if no DE immersion	100.00	%	0.08	Pass	+99.92
Regulation 28 GZ-based	28.3.3 Range of positive stability including DF	20.0	deg	73.5	Pass	+267.42
Regulation 28 GZ-based	28.3.3 Residual righting lever	0.100	m	0.406	Pass	+306.00

100% Loaded Vessel

Item Name	Quantity	Unit Mass tonne	Total Mass tonne	Long. Arm m	Vert. Arm m	Total FSM tonne.m
Lightship	1	131.830	131.830	16.540	3.360	0.000
Additional Weight of new Engine	1	5.498	5.498	12.200	1.240	0.000
Crew & Effects	1	4.000	4.000	20.000	4.100	0.000
Provisions	1	1.000	1.000	16.000	2.500	0.000
Ammunition	1	1.500	1.500	3.500	2.500	0.000
Liquid in systems	1	3.500	3.500	12.000	1.500	0.000
Tank 1 FRESH WATER AFT	100%	4.800	4.800	3.949	1.398	0.000
Tank 2 FRESH WATER FWD	100%	5.990	5.990	6.957	1.309	0.000
Tank 3 FUEL OIL AFT (Center)	100%	15.030	15.030	17.744	1.037	0.000
Tank 4 FUEL OIL FWD (Center)	100%	13.880	13.880	22.794	1.008	0.000
Tank 5 FUEL OIL (Stbd)	100%	6.370	6.370	15.899	2.633	0.000
Tank 6 FUEL OIL (Port)	100%	6.370	6.370	15.899	2.633	0.000
Tank 7 BLACK WATER	10%	0.240	0.240	25.787	0.270	1.298
Tank 8 LUB. OIL (Port)	100%	0.840	0.840	6.002	1.441	0.000
Tank 9 LUB. OIL (Stbd)	100%	0.840	0.840	6.002	1.441	0.000
Tank 10 DIRTY OIL (Centre Stbd)	10%	0.080	0.080	12.076	0.551	0.139
Total Loadcase			201.768	16.129	2.765	1.437
FS correction					0.007	
VCG fluid					2.772	

Draft Amidships m	1.880
Displacement t	201.8
Heel deg	0.0
Draft at FP m	1.831
Draft at AP m	1.928
Trim (+ve by stern) m	0.097
LCB from zero pt. (+ve fwd) m	16.125
LCF from zero pt. (+ve fwd) m	15.132
KB m	1.293
GMt corrected m	1.653
KML m	84.755
Immersion (TPc) tonne/cm	1.983
MTc tonne.m	4.610



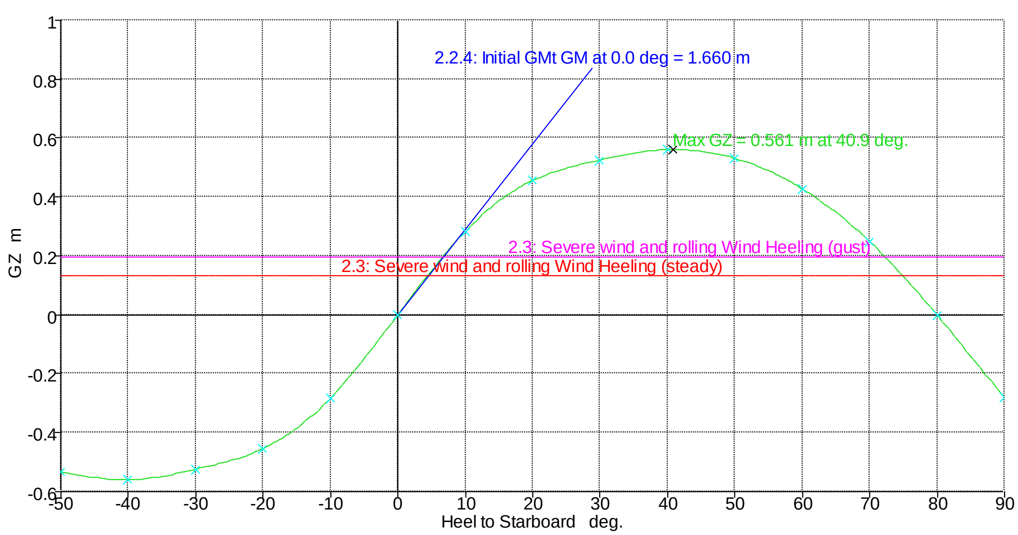
Heel to Starboard deg	0.0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	80.0	90.0
GZ m	0.001	0.287	0.507	0.630	0.701	0.673	0.573	0.411	0.183	-0.079
Area under GZ curve from zero heel m.deg	0.0012	1.4682	5.5161	11.2566	17.9701	24.9199	31.2016	36.1773	39.1952	39.7324
Displacement t	201.8	201.8	201.8	201.8	201.8	201.8	201.8	201.8	201.8	201.8
Draft at FP m	1.831	1.800	1.729	1.619	1.420	1.076	0.501	-0.632	-4.237	n/a
Draft at AP m	1.928	1.923	1.850	1.637	1.284	0.836	0.198	-0.962	-4.177	n/a
WL Length m	35.895	35.873	35.823	35.741	35.584	35.263	34.884	36.019	36.744	37.273
Beam max extents on WL m	6.895	6.800	6.258	6.123	5.757	5.744	5.518	4.800	4.122	3.725
Wetted Area m^2	246.500	246.551	238.495	234.420	237.194	242.546	246.088	245.058	243.701	243.693
Waterpl. Area m^2	198.300	198.084	186.495	180.905	172.258	157.927	147.054	133.583	122.381	116.287
Prismatic coeff. (Cp)	0.706	0.702	0.702	0.707	0.719	0.746	0.781	0.784	0.795	0.803
Block coeff. (Cb)	0.436	0.454	0.501	0.477	0.470	0.449	0.456	0.499	0.567	0.622
LCB from zero pt. (+ve fwd) m	16.125	16.124	16.124	16.127	16.135	16.137	16.136	16.138	16.128	16.118
LCF from zero pt. (+ve fwd) m	15.132	15.209	15.970	16.662	17.480	18.357	18.929	18.895	18.629	18.420
Max deck inclination deg	0.1549	10.0019	20.0008	30.0000	40.0003	50.0004	60.0003	70.0001	80.0000	90.0000
Trim angle (+ve by stern) deg	0.1549	0.1971	0.1931	0.0296	-0.2172	-0.3833	-0.4834	-0.5272	0.0950	n/a

Code	Criteria	Value	Units	Actual	Status	Margin %
267(85) Ch2 - General Criteria	2.3: IMO roll back angle	14.6	deg			
267(85) Ch2 - General Criteria	2.2.1: Area 0 to 30	3.1513	m.deg	11.2566	Pass	+257.20
267(85) Ch2 - General Criteria	2.2.1: Area 0 to 40	5.1566	m.deg	17.9701	Pass	+248.49
267(85) Ch2 - General Criteria	2.2.1: Area 30 to 40	1.7189	m.deg	6.7135	Pass	+290.57
267(85) Ch2 - General Criteria	2.2.2: Max GZ at 30 or greater	0.200	m	0.703	Pass	+251.50
267(85) Ch2 - General Criteria	2.2.3: Angle of maximum GZ	25.0	deg	41.8	Pass	+67.27
267(85) Ch2 - General Criteria	2.2.4: Initial GMt	0.150	m	1.653	Pass	+1002.00
267(85) Ch2 - General Criteria	2.3: Severe wind and rolling				Pass	
	Angle of steady heel shall not be greater than (<=)	16.0	deg	3.7	Pass	+76.61
	Angle of steady heel / Deck edge immersion angle shall not be greater than (<=)	80.00	%	10.89	Pass	+86.39
	Area1 / Area2 shall not be less than (>=)	100.00	%	426.59	Pass	+326.59
	Area 1 shall not be less than (>=)	0.0000	m.deg	17.0173	Pass	infinite
Regulation 28 GZ-based	28.3.2 Equi heel <= 25 or <= 30 if no DE immersion	100.00	%	0.07	Pass	+99.93
Regulation 28 GZ-based	28.3.3 Range of positive stability including DF	20.0	deg	87.1	Pass	+335.38
Regulation 28 GZ-based	28.3.3 Residual righting lever	0.100	m	0.506	Pass	+406.00

50% Loaded Vessel

Item Name	Quantity	Unit Mass tonne	Total Mass tonne	Long. Arm m	Vert. Arm m	Total FSM tonne.m
Lightship	1	131.830	131.830	16.540	3.360	0.000
Additional Weight of new Engine	1	5.498	5.498	12.200	1.240	0.000
Crew & Effects	1	4.000	4.000	20.000	4.100	0.000
Provisions	1	1.000	1.000	16.000	2.500	0.000
Ammunition	1	1.500	1.500	3.500	2.500	0.000
Liquid in systems	1	3.500	3.500	12.000	1.500	0.000
Tank 1 FRESH WATER AFT	50%	2.400	2.400	3.999	1.221	3.244
Tank 2 FRESH WATER FWD	50%	3.000	3.000	7.013	1.087	3.244
Tank 3 FUEL OIL AFT (Center)	50%	7.510	7.510	17.787	0.679	5.017
Tank 4 FUEL OIL FWD (Center)	50%	6.940	6.940	22.788	0.635	4.459
Tank 5 FUEL OIL (Stbd)	50%	3.180	3.180	15.893	1.889	0.442
Tank 6 FUEL OIL (Port)	50%	3.180	3.180	15.893	1.889	0.442
Tank 7 BLACK WATER	10%	0.240	0.240	25.787	0.270	1.298
Tank 8 LUB. OIL (Port)	50%	0.420	0.420	6.004	1.283	0.246
Tank 9 LUB. OIL (Stbd)	50%	0.420	0.420	6.004	1.283	0.246
Tank 10 DIRTY OIL (Centre Stbd)	10%	0.080	0.080	12.076	0.551	0.139
Total Loadcase			174.698	16.179	2.900	18.777
FS correction					0.107	
VCG fluid					3.007	

Draft Amidships m	1.738
Displacement t	174.7
Heel deg	0.0
Draft at FP m	1.666
Draft at AP m	1.809
Trim (+ve by stern) m	0.143
LCB from zero pt. (+ve fwd) m	16.171
LCF from zero pt. (+ve fwd) m	14.944
KB m	1.212
GMt corrected m	1.660
KML m	94.299
Immersion (TPc) tonne/cm	1.939
MTc tonne.m	4.445



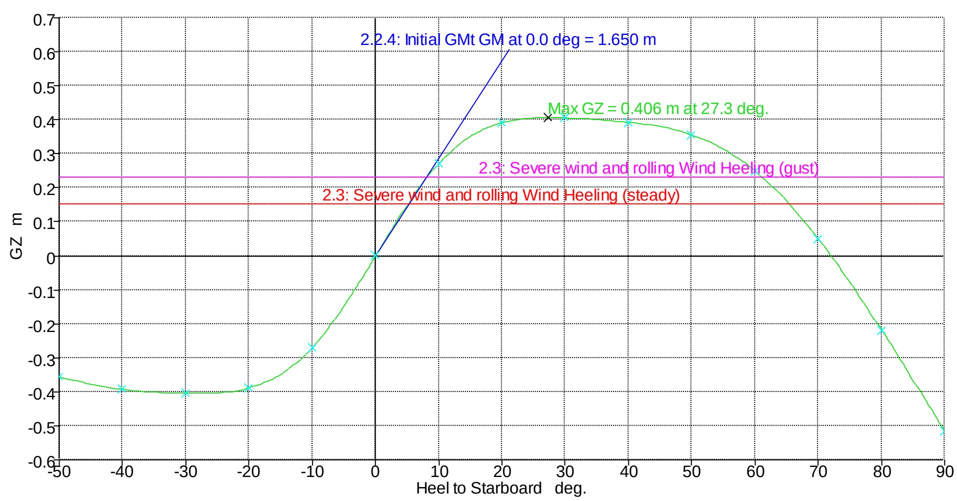
Heel to Starboard deg	0.0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	80.0	90.0
GZ m	0.001	0.284	0.455	0.525	0.561	0.532	0.429	0.249	-0.001	-0.282
Area under GZ curve from zero heel m.deg	0.0014	1.4793	5.2800	10.2331	15.6958	21.2196	26.0830	29.5362	30.8227	29.4227
Displacement t	174.7	174.7	174.7	174.7	174.7	174.7	174.7	174.7	174.7	174.7
Draft at FP m	1.667	1.636	1.578	1.479	1.284	0.940	0.342	-0.904	-4.870	n/a
Draft at AP m	1.809	1.799	1.686	1.428	1.011	0.433	-0.405	-1.924	-6.171	n/a
WL Length m	35.777	35.755	35.711	35.633	35.467	35.103	34.597	35.829	36.555	37.100
Beam max extents on WL m	6.872	6.698	6.060	5.928	5.818	5.786	5.605	4.605	3.956	3.611
Wetted Area m^2	234.464	233.223	222.915	218.836	219.967	225.112	228.265	225.504	224.170	224.097
Waterpl. Area m^2	193.948	191.633	177.237	171.623	170.700	157.410	146.205	129.820	118.877	112.805
Prismatic coeff. (Cp)	0.689	0.686	0.686	0.691	0.701	0.726	0.765	0.769	0.788	0.801
Block coeff. (Cb)	0.412	0.434	0.489	0.470	0.447	0.438	0.448	0.518	0.592	0.626
LCB from zero pt. (+ve fwd) m	16.172	16.171	16.174	16.184	16.192	16.198	16.200	16.202	16.196	16.182
LCF from zero pt. (+ve fwd) m	14.944	15.104	16.009	16.710	17.333	18.239	18.869	18.660	18.407	18.201
Max deck inclination deg	0.2282	10.0033	20.0006	30.0001	40.0012	50.0020	60.0018	70.0010	80.0002	90.0000
Trim angle (+ve by stern) deg	0.2282	0.2613	0.1729	-0.0816	-0.4363	-0.8096	-1.1923	-1.6297	-2.0765	n/a

Code	Criteria	Value	Units	Actual	Status	Margin %
267(85) Ch2 - General Criteria	2.3: IMO roll back angle	15.6	deg			
267(85) Ch2 - General Criteria	2.2.1: Area 0 to 30	3.1513	m.deg	10.2331	Pass	+224.73
267(85) Ch2 - General Criteria	2.2.1: Area 0 to 40	5.1566	m.deg	15.6958	Pass	+204.38
267(85) Ch2 - General Criteria	2.2.1: Area 30 to 40	1.7189	m.deg	5.4627	Pass	+217.80
267(85) Ch2 - General Criteria	2.2.2: Max GZ at 30 or greater	0.200	m	0.561	Pass	+180.50
267(85) Ch2 - General Criteria	2.2.3: Angle of maximum GZ	25.0	deg	40.9	Pass	+63.64
267(85) Ch2 - General Criteria	2.2.4: Initial GMt	0.150	m	1.660	Pass	+1006.67
267(85) Ch2 - General Criteria	2.3: Severe wind and rolling				Pass	
	Angle of steady heel shall not be greater than (<=)	16.0	deg	4.4	Pass	+72.78
	Angle of steady heel / Deck edge immersion angle shall not be greater than (<=)	80.00	%	11.59	Pass	+85.51
	Area1 / Area2 shall not be less than (>=)	100.00	%	252.72	Pass	+152.72
	Area 1 shall not be less than (>=)	0.0000	m.deg	11.9615	Pass	infinite
Regulation 28 GZ-based	28.3.2 Equi heel <= 25 or <= 30 if no DE immersion	100.00	%	0.08	Pass	+99.92
Regulation 28 GZ-based	28.3.3 Range of positive stability including DF	20.0	deg	80.0	Pass	+300.01
Regulation 28 GZ-based	28.3.3 Residual righting lever	0.100	m	0.455	Pass	+355.00

10% Loaded Vessel

Item Name	Quantity	Unit Mass tonne	Total Mass tonne	Long. Arm m	Vert. Arm m	Total FSM tonne.m
Lightship	1	131.830	131.830	16.540	3.360	0.000
Additional Weight of new Engine	1	5.498	5.498	12.200	1.240	0.000
Crew & Effects	1	4.000	4.000	20.000	4.100	0.000
Provisions	1	1.000	1.000	16.000	2.500	0.000
Ammunition	1	1.500	1.500	3.500	2.500	0.000
Liquid in systems	1	3.500	3.500	12.000	1.500	0.000
Tank 1 FRESH WATER AFT	10%	0.480	0.480	4.393	1.070	3.244
Tank 2 FRESH WATER FWD	10%	0.600	0.600	7.437	0.890	3.244
Tank 3 FUEL OIL AFT (Center)	10%	1.500	1.500	18.116	0.373	5.017
Tank 4 FUEL OIL FWD (Center)	10%	1.390	1.390	22.770	0.312	4.459
Tank 5 FUEL OIL (Stbd)	10%	0.640	0.640	15.892	1.175	0.442
Tank 6 FUEL OIL (Port)	10%	0.640	0.640	15.892	1.175	0.442
Tank 7 BLACK WATER	10%	0.240	0.240	25.787	0.270	1.298
Tank 8 LUB. OIL (Port)	10%	0.080	0.080	6.023	1.141	0.246
Tank 9 LUB. OIL (Stbd)	10%	0.080	0.080	6.023	1.141	0.246
Tank 10 DIRTY OIL (Centre Stbd)	10%	0.080	0.080	12.076	0.551	0.139
Total Loadcase			153.058	16.233	3.146	18.777
FS correction					0.123	
VCG fluid					3.269	

Draft Amidships m	1.621
Displacement t	153.1
Heel deg	0.0
Draft at FP m	1.526
Draft at AP m	1.716
Trim (+ve by stern) m	0.189
LCB from zero pt. (+ve fwd) m	16.221
LCF from zero pt. (+ve fwd) m	14.776
KB m	1.144
GMt corrected m	1.650
KML m	103.932
Immersion (TPc) tonne/cm	1.899
MTc tonne.m	4.294



Heel to Starboard deg	0.0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	80.0	90.0
GZ m	0.001	0.272	0.390	0.405	0.393	0.357	0.250	0.051	-0.218	-0.514
Area under GZ curve from zero heel m.deg	0.0015	1.4461	4.8807	8.9059	12.9078	16.6916	19.7975	21.3720	20.5728	16.9207
Displacement t	153.1	153.1	153.0	153.1	153.1	153.1	153.1	153.1	153.1	153.1
Draft at FP m	1.527	1.498	1.452	1.362	1.174	0.829	0.213	-1.141	-5.429	n/a
Draft at AP m	1.716	1.697	1.547	1.251	0.790	0.112	-0.888	-2.693	-7.757	n/a
WL Length m	35.672	35.650	35.613	35.536	35.362	34.954	34.331	35.652	36.380	36.935
Beam max extents on WL m	6.840	6.472	5.887	5.760	5.775	5.763	5.674	4.445	3.812	3.530
Wetted Area m^2	224.430	220.222	209.656	205.655	206.395	210.851	213.180	209.495	208.137	208.001
Waterpl. Area m^2	189.902	183.735	169.102	163.582	165.372	156.355	144.293	126.566	115.797	109.728
Prismatic coeff. (Cp)	0.673	0.671	0.671	0.676	0.684	0.707	0.749	0.755	0.780	0.799
Block coeff. (Cb)	0.391	0.426	0.478	0.463	0.432	0.428	0.435	0.529	0.600	0.616
LCB from zero pt. (+ve fwd) m	16.222	16.222	16.228	16.240	16.253	16.265	16.268	16.271	16.263	16.248
LCF from zero pt. (+ve fwd) m	14.776	15.117	16.047	16.752	17.356	18.131	18.773	18.449	18.203	18.003
Max deck inclination deg	0.3021	10.0049	20.0005	30.0004	40.0023	50.0040	60.0039	70.0023	80.0006	90.0000
Trim angle (+ve by stern) deg	0.3021	0.3181	0.1508	-0.1777	-0.6136	-1.1445	-1.7587	-2.4772	-3.7123	n/a

Code	Criteria	Value	Units	Actual	Status	Margin %
267(85) Ch2 - General Criteria	2.3: IMO roll back angle	16.8	deg			
267(85) Ch2 - General Criteria	2.2.1: Area 0 to 30	3.1513	m.deg	8.9059	Pass	+182.61
267(85) Ch2 - General Criteria	2.2.1: Area 0 to 40	5.1566	m.deg	12.9078	Pass	+150.32
267(85) Ch2 - General Criteria	2.2.1: Area 30 to 40	1.7189	m.deg	4.0019	Pass	+132.82
267(85) Ch2 - General Criteria	2.2.2: Max GZ at 30 or greater	0.200	m	0.405	Pass	+102.50
267(85) Ch2 - General Criteria	2.2.3: Angle of maximum GZ	25.0	deg	27.3	Pass	+9.09
267(85) Ch2 - General Criteria	2.2.4: Initial GMt	0.150	m	1.650	Pass	+1000.00
267(85) Ch2 - General Criteria	2.3: Severe wind and rolling				Pass	
	Angle of steady heel shall not be greater than (<=)	16.0	deg	5.2	Pass	+67.64
	Angle of steady heel / Deck edge immersion angle shall not be greater than (<=)	80.00	%	12.84	Pass	+83.95
	Area1 / Area2 shall not be less than (>=)	100.00	%	111.14	Pass	+11.14
	Area 1 shall not be less than (>=)	0.0000	m.deg	6.0589	Pass	infinite
Regulation 28 GZ-based	28.3.2 Equi heel <= 25 or <= 30 if no DE immersion	100.00	%	0.09	Pass	+99.91
Regulation 28 GZ-based	28.3.3 Range of positive stability including DF	20.0	deg	72.1	Pass	+260.32
Regulation 28 GZ-based	28.3.3 Residual righting lever	0.100	m	0.390	Pass	+290.00

Hydrostatics

Fixed Trim =-1 m (+ve by stern)

Specific gravity = 1.025; (Density = 1.025 tonne/m^3)

Draft Amidships m	Displacement t	Draft at FP m	Draft at AP m	Trim (+ve by stern) m	Wetted Area m^2	Block coeff. (Cb)	LCB from zero pt. (+ve fwd) m	VCB m	LCF from zero pt. (+ve fwd) m	KB m	BMt m	GMt m	KMt m	Immersion (TPc) tonne/cm	MTc tonne.m
0.100	2.657	0.600	-0.400	-1.000	20.243	0.243	26.966	0.241	25.705	0.241	0.827	-0.993	1.068	0.178	0.080
0.150	3.636	0.650	-0.350	-1.000	24.276	0.235	26.575	0.267	25.345	0.267	1.013	-0.771	1.279	0.214	0.103
0.200	4.803	0.700	-0.300	-1.000	28.524	0.228	26.234	0.294	25.014	0.294	1.200	-0.546	1.494	0.253	0.129
0.250	6.167	0.750	-0.250	-1.000	32.941	0.223	25.929	0.321	24.707	0.321	1.379	-0.331	1.700	0.293	0.158
0.300	7.733	0.800	-0.200	-1.000	37.462	0.223	25.651	0.349	24.415	0.349	1.535	-0.140	1.884	0.334	0.191
0.350	9.505	0.850	-0.150	-1.000	42.075	0.226	25.393	0.378	24.138	0.378	1.666	0.027	2.044	0.375	0.227
0.400	11.49	0.900	-0.100	-1.000	46.807	0.230	25.152	0.406	23.864	0.406	1.783	0.179	2.189	0.418	0.268
0.450	13.68	0.950	-0.050	-1.000	51.639	0.234	24.924	0.435	23.603	0.435	1.891	0.323	2.326	0.461	0.313
0.500	16.10	1.000	0.000	-1.000	56.581	0.238	24.706	0.465	23.350	0.465	1.995	0.461	2.459	0.505	0.361
0.550	18.73	1.050	0.050	-1.000	61.631	0.242	24.497	0.494	23.104	0.494	2.093	0.595	2.586	0.550	0.414
0.600	21.59	1.100	0.100	-1.000	66.779	0.246	24.296	0.523	22.862	0.523	2.183	0.720	2.706	0.595	0.471
0.650	24.69	1.150	0.150	-1.000	72.027	0.250	24.101	0.553	22.621	0.553	2.266	0.837	2.818	0.642	0.534
0.700	28.01	1.200	0.200	-1.000	77.390	0.254	23.910	0.582	22.377	0.582	2.342	0.948	2.923	0.689	0.602
0.750	31.58	1.250	0.250	-1.000	82.897	0.258	23.723	0.612	22.129	0.612	2.415	1.056	3.026	0.738	0.677
0.800	35.39	1.300	0.300	-1.000	88.569	0.262	23.537	0.642	21.874	0.642	2.487	1.164	3.128	0.787	0.759
0.850	39.45	1.350	0.350	-1.000	94.425	0.265	23.352	0.672	21.612	0.672	2.561	1.272	3.231	0.839	0.849
0.900	43.78	1.400	0.400	-1.000	100.477	0.269	23.167	0.701	21.340	0.701	2.635	1.382	3.336	0.892	0.949
0.950	48.37	1.450	0.450	-1.000	106.723	0.273	22.980	0.731	21.062	0.731	2.710	1.491	3.440	0.947	1.058
1.000	53.25	1.500	0.500	-1.000	113.152	0.276	22.791	0.761	20.775	0.761	2.783	1.599	3.543	1.003	1.177
1.050	58.41	1.550	0.550	-1.000	119.761	0.279	22.600	0.791	20.480	0.791	2.853	1.705	3.643	1.061	1.308
1.100	63.86	1.600	0.600	-1.000	126.555	0.282	22.406	0.821	20.175	0.821	2.921	1.808	3.741	1.120	1.451
1.150	69.61	1.650	0.650	-1.000	133.545	0.284	22.209	0.851	19.862	0.851	2.988	1.910	3.838	1.181	1.608
1.200	75.67	1.700	0.700	-1.000	140.742	0.287	22.008	0.881	19.539	0.881	3.055	2.013	3.935	1.244	1.779
1.250	82.04	1.750	0.750	-1.000	148.163	0.289	21.803	0.911	19.207	0.911	3.123	2.116	4.032	1.308	1.965
1.300	88.75	1.800	0.800	-1.000	155.803	0.291	21.594	0.941	18.866	0.941	3.193	2.222	4.132	1.374	2.168
1.350	95.78	1.850	0.850	-1.000	163.553	0.293	21.381	0.970	18.530	0.970	3.267	2.331	4.236	1.441	2.380
1.400	103.1	1.900	0.900	-1.000	171.410	0.295	21.165	1.000	18.197	1.000	3.344	2.444	4.343	1.508	2.604
1.450	110.9	1.950	0.950	-1.000	179.475	0.299	20.947	1.030	17.857	1.030	3.426	2.562	4.454	1.577	2.845
1.500	118.9	2.000	1.000	-1.000	187.862	0.303	20.725	1.060	17.498	1.060	3.514	2.687	4.573	1.648	3.111
1.550	127.3	2.050	1.050	-1.000	195.958	0.307	20.501	1.089	17.166	1.089	3.559	2.767	4.647	1.715	3.383
1.600	136.1	2.100	1.100	-1.000	203.963	0.311	20.276	1.119	16.846	1.119	3.576	2.820	4.693	1.780	3.667
1.650	145.1	2.150	1.150	-1.000	212.071	0.316	20.052	1.148	16.524	1.148	3.583	2.863	4.730	1.844	3.969
1.700	154.5	2.200	1.200	-1.000	220.316	0.320	19.828	1.177	16.200	1.177	3.585	2.900	4.761	1.909	4.290

Draft Amidships m	Displacement t	Draft at FP m	Draft at AP m	Trim (+ve by stern) m	Wetted Area m^2	Block coeff. (Cb)	LCB from zero pt. (+ve fwd) m	VCB m	LCF from zero pt. (+ve fwd) m	KB m	BMt m	GMt m	KMt m	Immersion (TPc) tonne/cm	MTc tonne.m
1.750	164.2	2.250	1.250	-1.000	228.698	0.324	19.604	1.207	15.874	1.207	3.581	2.932	4.787	1.974	4.633
1.800	174.2	2.300	1.300	-1.000	237.118	0.327	19.380	1.236	15.556	1.236	3.573	2.959	4.807	2.039	4.988
1.850	184.5	2.350	1.350	-1.000	241.716	0.336	19.167	1.265	15.574	1.265	3.468	2.889	4.731	2.060	5.059
1.900	194.8	2.400	1.400	-1.000	246.178	0.344	18.978	1.294	15.612	1.294	3.366	2.821	4.658	2.079	5.116
1.950	205.3	2.450	1.450	-1.000	250.633	0.352	18.808	1.323	15.656	1.323	3.270	2.759	4.591	2.098	5.170
2.000	215.8	2.500	1.500	-1.000	255.052	0.361	18.655	1.351	15.704	1.351	3.177	2.699	4.527	2.116	5.222

Fixed Trim =-0.75 m (+ve by stern)

Draft Amidships m	Displacement t	Draft at FP m	Draft at AP m	Trim (+ve by stern) m	Wetted Area m^2	Block coeff. (Cb)	LCB from zero pt. (+ve fwd) m	VCB m	LCF from zero pt. (+ve fwd) m	KB m	BMt m	GMt m	KMt m	Immersion (TPc) tonne/cm	MTc tonne.m
0.100	1.788	0.475	-0.275	-0.750	16.727	0.233	26.601	0.193	25.468	0.193	0.760	-1.038	0.953	0.148	0.065
0.150	2.621	0.525	-0.225	-0.750	20.796	0.225	26.177	0.220	25.091	0.220	0.965	-0.797	1.185	0.186	0.087
0.200	3.647	0.575	-0.175	-0.750	25.107	0.220	25.821	0.249	24.747	0.249	1.170	-0.555	1.419	0.225	0.112
0.250	4.874	0.625	-0.125	-0.750	29.648	0.217	25.507	0.278	24.422	0.278	1.373	-0.317	1.650	0.267	0.141
0.300	6.314	0.675	-0.075	-0.750	34.352	0.218	25.224	0.308	24.117	0.308	1.554	-0.100	1.861	0.310	0.175
0.350	7.971	0.725	-0.025	-0.750	39.169	0.222	24.962	0.337	23.824	0.337	1.706	0.087	2.043	0.353	0.212
0.400	9.849	0.775	0.025	-0.750	44.089	0.228	24.717	0.368	23.540	0.368	1.836	0.252	2.203	0.398	0.255
0.450	11.95	0.825	0.075	-0.750	49.102	0.233	24.485	0.398	23.268	0.398	1.953	0.404	2.350	0.443	0.301
0.500	14.28	0.875	0.125	-0.750	54.229	0.238	24.264	0.428	23.006	0.428	2.062	0.549	2.490	0.489	0.352
0.550	16.85	0.925	0.175	-0.750	59.482	0.242	24.052	0.459	22.747	0.459	2.166	0.687	2.624	0.536	0.408
0.600	19.65	0.975	0.225	-0.750	64.860	0.247	23.848	0.489	22.491	0.489	2.263	0.819	2.752	0.585	0.469
0.650	22.69	1.025	0.275	-0.750	70.369	0.251	23.648	0.520	22.236	0.520	2.353	0.944	2.873	0.634	0.536
0.700	25.99	1.075	0.325	-0.750	76.026	0.256	23.452	0.551	21.975	0.551	2.437	1.063	2.988	0.684	0.611
0.750	29.54	1.125	0.375	-0.750	81.856	0.260	23.258	0.581	21.706	0.581	2.519	1.179	3.100	0.736	0.693
0.800	33.35	1.175	0.425	-0.750	87.886	0.265	23.065	0.612	21.430	0.612	2.600	1.295	3.212	0.790	0.784
0.850	37.44	1.225	0.475	-0.750	94.139	0.269	22.871	0.643	21.142	0.643	2.683	1.413	3.326	0.845	0.886
0.900	41.81	1.275	0.525	-0.750	100.628	0.273	22.674	0.674	20.843	0.674	2.769	1.534	3.442	0.903	0.999
0.950	46.47	1.325	0.575	-0.750	107.359	0.276	22.475	0.705	20.535	0.705	2.856	1.657	3.561	0.963	1.123
1.000	51.44	1.375	0.625	-0.750	114.327	0.279	22.272	0.736	20.215	0.736	2.944	1.779	3.679	1.025	1.261
1.050	56.72	1.425	0.675	-0.750	121.525	0.282	22.065	0.767	19.886	0.767	3.030	1.901	3.796	1.089	1.414
1.100	62.33	1.475	0.725	-0.750	128.961	0.285	21.853	0.798	19.544	0.798	3.114	2.021	3.912	1.155	1.583
1.150	68.28	1.525	0.775	-0.750	136.646	0.287	21.637	0.829	19.190	0.829	3.199	2.141	4.028	1.223	1.769
1.200	74.56	1.575	0.825	-0.750	144.592	0.290	21.415	0.860	18.825	0.860	3.285	2.263	4.145	1.293	1.973
1.250	81.21	1.625	0.875	-0.750	152.686	0.292	21.188	0.891	18.461	0.891	3.374	2.388	4.265	1.365	2.191
1.300	88.21	1.675	0.925	-0.750	160.880	0.295	20.957	0.922	18.103	0.922	3.467	2.517	4.389	1.436	2.420
1.350	95.57	1.725	0.975	-0.750	169.292	0.297	20.723	0.953	17.742	0.953	3.568	2.653	4.521	1.510	2.665
1.400	103.3	1.775	1.025	-0.750	178.204	0.300	20.485	0.984	17.353	0.984	3.693	2.814	4.676	1.588	2.940
1.450	111.4	1.825	1.075	-0.750	187.206	0.304	20.242	1.015	16.962	1.015	3.798	2.956	4.813	1.666	3.239
1.500	120.0	1.875	1.125	-0.750	195.817	0.308	19.996	1.046	16.602	1.046	3.840	3.033	4.885	1.738	3.544
1.550	128.8	1.925	1.175	-0.750	204.497	0.311	19.750	1.077	16.243	1.077	3.862	3.091	4.938	1.810	3.870
1.600	138.1	1.975	1.225	-0.750	213.322	0.315	19.504	1.108	15.880	1.108	3.872	3.137	4.979	1.881	4.221
1.650	147.6	2.025	1.275	-0.750	222.303	0.320	19.256	1.138	15.515	1.138	3.874	3.175	5.011	1.953	4.597
1.700	157.5	2.075	1.325	-0.750	229.136	0.327	19.012	1.169	15.341	1.169	3.809	3.145	4.977	2.000	4.832
1.750	167.6	2.125	1.375	-0.750	233.540	0.335	18.793	1.199	15.376	1.199	3.669	3.040	4.867	2.019	4.891
1.800	177.7	2.175	1.425	-0.750	237.916	0.344	18.600	1.229	15.413	1.229	3.540	2.945	4.768	2.037	4.947
1.850	188.0	2.225	1.475	-0.750	242.273	0.353	18.427	1.258	15.454	1.258	3.420	2.858	4.678	2.055	5.001
1.900	198.3	2.275	1.525	-0.750	246.651	0.362	18.274	1.288	15.500	1.288	3.311	2.782	4.598	2.072	5.052
1.950	208.7	2.325	1.575	-0.750	251.054	0.371	18.137	1.317	15.551	1.317	3.212	2.715	4.528	2.089	5.103
2.000	219.2	2.375	1.625	-0.750	255.438	0.380	18.014	1.346	15.602	1.346	3.118	2.653	4.464	2.106	5.153

Fixed Trim =-0.5 m (+ve by stern)

Draft Amidships m	Displacement t	Draft at FP m	Draft at AP m	Trim (+ve by stern) m	Wetted Area m^2	Block coeff. (Cb)	LCB from zero pt. (+ve fwd) m	VCB m	LCF from zero pt. (+ve fwd) m	KB m	BMt m	GMt m	KMt m	Immersion (TPc) tonne/cm	MTc tonne.m
0.100	1.101	0.350	-0.150	-0.500	13.222	0.218	26.082	0.147	25.145	0.147	0.694	-1.083	0.841	0.119	0.050
0.150	1.788	0.400	-0.100	-0.500	17.333	0.215	25.642	0.176	24.757	0.176	0.926	-0.815	1.102	0.156	0.070
0.200	2.670	0.450	-0.050	-0.500	21.707	0.215	25.288	0.207	24.408	0.207	1.151	-0.555	1.357	0.197	0.095
0.250	3.760	0.500	0.000	-0.500	26.357	0.215	24.982	0.238	24.076	0.238	1.373	-0.298	1.610	0.240	0.124
0.300	5.071	0.550	0.050	-0.500	31.243	0.216	24.705	0.269	23.759	0.269	1.580	-0.055	1.849	0.285	0.157
0.350	6.611	0.600	0.100	-0.500	36.293	0.221	24.447	0.301	23.449	0.301	1.757	0.157	2.057	0.331	0.197
0.400	8.385	0.650	0.150	-0.500	41.431	0.228	24.204	0.332	23.159	0.332	1.903	0.338	2.235	0.378	0.240
0.450	10.40	0.700	0.200	-0.500	46.663	0.235	23.974	0.364	22.880	0.364	2.031	0.501	2.394	0.426	0.289
0.500	12.65	0.750	0.250	-0.500	52.015	0.240	23.755	0.396	22.606	0.396	2.147	0.652	2.543	0.475	0.343
0.550	15.14	0.800	0.300	-0.500	57.506	0.246	23.543	0.427	22.336	0.427	2.256	0.796	2.684	0.524	0.402
0.600	17.89	0.850	0.350	-0.500	63.153	0.250	23.336	0.459	22.066	0.459	2.360	0.934	2.819	0.575	0.468
0.650	20.90	0.900	0.400	-0.500	68.973	0.255	23.133	0.491	21.791	0.491	2.458	1.067	2.949	0.628	0.542
0.700	24.17	0.950	0.450	-0.500	74.987	0.260	22.932	0.523	21.509	0.523	2.552	1.196	3.075	0.682	0.624
0.750	27.73	1.000	0.500	-0.500	81.222	0.265	22.731	0.555	21.216	0.555	2.644	1.323	3.199	0.738	0.716
0.800	31.56	1.050	0.550	-0.500	87.707	0.270	22.528	0.587	20.909	0.587	2.738	1.451	3.325	0.797	0.820
0.850	35.70	1.100	0.600	-0.500	94.467	0.274	22.322	0.619	20.588	0.619	2.835	1.583	3.454	0.858	0.937
0.900	40.14	1.150	0.650	-0.500	101.518	0.278	22.111	0.652	20.254	0.652	2.936	1.719	3.587	0.921	1.067
0.950	44.92	1.200	0.700	-0.500	108.875	0.281	21.895	0.684	19.905	0.684	3.040	1.859	3.724	0.988	1.213
1.000	50.03	1.250	0.750	-0.500	116.539	0.284	21.673	0.716	19.542	0.716	3.148	2.002	3.863	1.057	1.378
1.050	55.49	1.300	0.800	-0.500	124.513	0.287	21.444	0.748	19.165	0.748	3.256	2.146	4.004	1.129	1.561
1.100	61.32	1.350	0.850	-0.500	132.805	0.290	21.209	0.781	18.773	0.781	3.365	2.291	4.146	1.204	1.766
1.150	67.53	1.400	0.900	-0.500	141.313	0.293	20.966	0.813	18.377	0.813	3.477	2.438	4.290	1.281	1.989
1.200	74.13	1.450	0.950	-0.500	149.914	0.296	20.719	0.846	17.990	0.846	3.591	2.588	4.436	1.358	2.224
1.250	81.11	1.500	1.000	-0.500	158.689	0.298	20.467	0.878	17.605	0.878	3.709	2.742	4.587	1.436	2.475
1.300	88.49	1.550	1.050	-0.500	167.932	0.301	20.211	0.911	17.198	0.911	3.851	2.920	4.761	1.518	2.754
1.350	96.31	1.600	1.100	-0.500	178.010	0.302	19.948	0.943	16.742	0.943	4.043	3.149	4.986	1.609	3.077
1.400	104.6	1.650	1.150	-0.500	187.543	0.302	19.678	0.976	16.325	0.976	4.142	3.283	5.117	1.692	3.411
1.450	113.2	1.700	1.200	-0.500	196.993	0.306	19.406	1.008	15.918	1.008	4.192	3.370	5.200	1.773	3.769
1.500	122.3	1.750	1.250	-0.500	206.581	0.311	19.132	1.041	15.507	1.041	4.221	3.435	5.261	1.854	4.158
1.550	131.8	1.800	1.300	-0.500	216.204	0.316	18.856	1.073	15.103	1.073	4.233	3.483	5.305	1.933	4.569
1.600	141.5	1.850	1.350	-0.500	220.931	0.327	18.598	1.105	15.123	1.105	4.071	3.357	5.175	1.957	4.651
1.650	151.3	1.900	1.400	-0.500	225.377	0.336	18.374	1.136	15.165	1.136	3.906	3.226	5.042	1.977	4.712
1.700	161.3	1.950	1.450	-0.500	229.782	0.346	18.178	1.167	15.208	1.167	3.755	3.109	4.921	1.996	4.772
1.750	171.3	2.000	1.500	-0.500	234.148	0.355	18.005	1.197	15.252	1.197	3.614	3.001	4.811	2.014	4.828
1.800	181.4	2.050	1.550	-0.500	238.477	0.364	17.853	1.227	15.298	1.227	3.482	2.901	4.709	2.031	4.881
1.850	191.6	2.100	1.600	-0.500	242.786	0.374	17.718	1.257	15.346	1.257	3.360	2.810	4.617	2.047	4.932
1.900	201.9	2.150	1.650	-0.500	247.112	0.383	17.599	1.287	15.397	1.287	3.249	2.730	4.535	2.063	4.982
1.950	212.2	2.200	1.700	-0.500	251.479	0.392	17.493	1.316	15.450	1.316	3.150	2.662	4.466	2.079	5.032
2.000	222.7	2.250	1.750	-0.500	255.838	0.401	17.398	1.346	15.504	1.346	3.057	2.600	4.403	2.095	5.081

Fixed Trim =-0.25 m (+ve by stern)

Draft Amidships m	Displacement t	Draft at FP m	Draft at AP m	Trim (+ve by stern) m	Wetted Area m^2	Block coeff. (Cb)	LCB from zero pt. (+ve fwd) m	VCB m	LCF from zero pt. (+ve fwd) m	KB m	BMt m	GMt m	KMt m	Immersion (TPc) tonne/cm	MTc tonne.m
0.100	0.5961	0.225	-0.025	-0.250	9.747	0.203	25.279	0.104	24.656	0.104	0.638	-1.119	0.742	0.089	0.034
0.150	1.135	0.275	0.025	-0.250	13.896	0.209	24.891	0.136	24.298	0.136	0.903	-0.819	1.039	0.127	0.053
0.200	1.873	0.325	0.075	-0.250	18.345	0.213	24.588	0.169	23.967	0.169	1.150	-0.538	1.318	0.169	0.077
0.250	2.825	0.375	0.125	-0.250	23.106	0.216	24.322	0.202	23.642	0.202	1.388	-0.264	1.590	0.213	0.105
0.300	4.006	0.425	0.175	-0.250	28.172	0.219	24.072	0.235	23.319	0.235	1.618	0.000	1.853	0.260	0.140
0.350	5.429	0.475	0.225	-0.250	33.461	0.225	23.832	0.268	23.006	0.268	1.820	0.237	2.088	0.309	0.180
0.400	7.100	0.525	0.275	-0.250	38.854	0.232	23.601	0.302	22.711	0.302	1.987	0.440	2.289	0.359	0.225
0.450	9.021	0.575	0.325	-0.250	44.351	0.240	23.380	0.335	22.425	0.335	2.128	0.615	2.463	0.410	0.276
0.500	11.20	0.625	0.375	-0.250	49.977	0.247	23.166	0.368	22.142	0.368	2.252	0.774	2.620	0.461	0.333
0.550	13.64	0.675	0.425	-0.250	55.761	0.252	22.958	0.401	21.859	0.401	2.367	0.924	2.769	0.514	0.398
0.600	16.34	0.725	0.475	-0.250	61.739	0.257	22.751	0.434	21.570	0.434	2.478	1.069	2.912	0.569	0.470
0.650	19.33	0.775	0.525	-0.250	67.944	0.262	22.546	0.468	21.272	0.468	2.585	1.211	3.053	0.625	0.552
0.700	22.60	0.825	0.575	-0.250	74.409	0.268	22.338	0.501	20.961	0.501	2.692	1.352	3.193	0.684	0.645
0.750	26.18	0.875	0.625	-0.250	81.166	0.273	22.128	0.534	20.633	0.534	2.800	1.495	3.334	0.746	0.751
0.800	30.07	0.925	0.675	-0.250	88.246	0.277	21.912	0.568	20.286	0.568	2.912	1.642	3.479	0.811	0.872
0.850	34.29	0.975	0.725	-0.250	95.679	0.281	21.689	0.601	19.921	0.601	3.029	1.794	3.630	0.879	1.009
0.900	38.86	1.025	0.775	-0.250	103.488	0.285	21.458	0.635	19.536	0.635	3.153	1.954	3.788	0.950	1.166
0.950	43.80	1.075	0.825	-0.250	111.695	0.288	21.219	0.669	19.132	0.669	3.283	2.120	3.952	1.026	1.344
1.000	49.13	1.125	0.875	-0.250	120.318	0.292	20.969	0.703	18.709	0.703	3.420	2.292	4.123	1.105	1.547
1.050	54.86	1.175	0.925	-0.250	129.289	0.295	20.710	0.737	18.274	0.737	3.563	2.470	4.299	1.187	1.773
1.100	61.00	1.225	0.975	-0.250	138.391	0.298	20.443	0.771	17.851	0.771	3.708	2.652	4.479	1.270	2.014
1.150	67.56	1.275	1.025	-0.250	147.638	0.301	20.171	0.805	17.435	0.805	3.855	2.834	4.660	1.354	2.272
1.200	74.55	1.325	1.075	-0.250	157.280	0.304	19.894	0.839	17.004	0.839	4.016	3.032	4.856	1.442	2.557
1.250	81.99	1.375	1.125	-0.250	167.947	0.300	19.610	0.874	16.512	0.874	4.250	3.302	5.124	1.540	2.893
1.300	89.95	1.425	1.175	-0.250	178.959	0.297	19.313	0.908	16.010	0.908	4.472	3.561	5.380	1.640	3.267
1.350	98.39	1.475	1.225	-0.250	189.546	0.301	19.009	0.943	15.534	0.943	4.587	3.712	5.529	1.734	3.666
1.400	107.3	1.525	1.275	-0.250	200.177	0.306	18.701	0.977	15.059	0.977	4.653	3.814	5.630	1.827	4.104
1.450	116.6	1.575	1.325	-0.250	208.018	0.313	18.395	1.011	14.831	1.011	4.583	3.781	5.594	1.887	4.384
1.500	126.1	1.625	1.375	-0.250	212.606	0.327	18.129	1.045	14.880	1.045	4.372	3.605	5.416	1.910	4.454
1.550	135.7	1.675	1.425	-0.250	217.120	0.339	17.901	1.077	14.930	1.077	4.177	3.445	5.254	1.932	4.520
1.600	145.4	1.725	1.475	-0.250	221.580	0.349	17.704	1.109	14.981	1.109	4.000	3.301	5.109	1.952	4.583
1.650	155.2	1.775	1.525	-0.250	225.994	0.359	17.533	1.140	15.032	1.140	3.837	3.170	4.977	1.972	4.643
1.700	165.1	1.825	1.575	-0.250	230.362	0.369	17.385	1.171	15.084	1.171	3.686	3.051	4.857	1.990	4.700
1.750	175.1	1.875	1.625	-0.250	234.695	0.379	17.255	1.202	15.137	1.202	3.545	2.941	4.746	2.006	4.755
1.800	185.2	1.925	1.675	-0.250	239.000	0.388	17.141	1.232	15.189	1.232	3.414	2.842	4.646	2.022	4.807
1.850	195.4	1.975	1.725	-0.250	243.287	0.398	17.041	1.262	15.242	1.262	3.294	2.751	4.555	2.038	4.858
1.900	205.6	2.025	1.775	-0.250	247.579	0.407	16.953	1.291	15.295	1.291	3.183	2.671	4.474	2.053	4.909
1.950	215.9	2.075	1.825	-0.250	251.916	0.416	16.875	1.321	15.351	1.321	3.085	2.603	4.406	2.069	4.958
2.000	226.3	2.125	1.875	-0.250	256.256	0.425	16.807	1.350	15.408	1.350	2.995	2.542	4.344	2.084	5.007

Fixed Trim = 0 m (+ve by stern)

Draft Amidships m	Displacement t	Draft at FP m	Draft at AP m	Trim (+ve by stern) m	Wetted Area m^2	Block coeff. (Cb)	LCB from zero pt. (+ve fwd) m	VCB m	LCF from zero pt. (+ve fwd) m	KB m	BMt m	GMt m	KMt m	Immersion (TPc) tonne/cm	MTc tonne.m
0.100	0.2687	0.100	0.100	0.000	6.384	0.211	23.958	0.068	23.835	0.068	0.615	-1.127	0.683	0.059	0.019
0.150	0.6595	0.150	0.150	0.000	10.514	0.219	23.821	0.103	23.628	0.103	0.909	-0.798	1.012	0.098	0.036
0.200	1.254	0.200	0.200	0.000	15.039	0.226	23.667	0.138	23.374	0.138	1.176	-0.496	1.314	0.140	0.058
0.250	2.069	0.250	0.250	0.000	19.930	0.230	23.493	0.173	23.086	0.173	1.429	-0.208	1.602	0.186	0.086
0.300	3.122	0.300	0.300	0.000	25.190	0.232	23.302	0.208	22.776	0.208	1.675	0.073	1.883	0.236	0.120
0.350	4.429	0.350	0.350	0.000	30.719	0.237	23.100	0.242	22.470	0.242	1.901	0.333	2.143	0.288	0.161
0.400	5.999	0.400	0.400	0.000	36.400	0.244	22.895	0.277	22.177	0.277	2.090	0.558	2.368	0.341	0.209
0.450	7.837	0.450	0.450	0.000	42.215	0.252	22.692	0.312	21.888	0.312	2.246	0.749	2.559	0.395	0.263
0.500	9.949	0.500	0.500	0.000	48.179	0.259	22.490	0.347	21.596	0.347	2.380	0.917	2.727	0.450	0.325
0.550	12.34	0.550	0.550	0.000	54.334	0.265	22.287	0.381	21.296	0.381	2.503	1.075	2.885	0.507	0.396
0.600	15.02	0.600	0.600	0.000	60.737	0.270	22.082	0.416	20.984	0.416	2.623	1.229	3.039	0.566	0.477
0.650	18.01	0.650	0.650	0.000	67.445	0.275	21.873	0.451	20.654	0.451	2.744	1.384	3.194	0.628	0.570
0.700	21.31	0.700	0.700	0.000	74.507	0.280	21.656	0.486	20.302	0.486	2.869	1.544	3.354	0.693	0.678
0.750	24.95	0.750	0.750	0.000	81.967	0.284	21.431	0.521	19.926	0.521	3.000	1.711	3.521	0.762	0.804
0.800	28.94	0.800	0.800	0.000	89.867	0.289	21.196	0.556	19.523	0.556	3.140	1.886	3.696	0.835	0.949
0.850	33.31	0.850	0.850	0.000	98.246	0.292	20.948	0.591	19.092	0.591	3.290	2.072	3.882	0.913	1.119
0.900	38.08	0.900	0.900	0.000	107.142	0.296	20.686	0.627	18.634	0.627	3.452	2.269	4.079	0.996	1.316
0.950	43.28	0.950	0.950	0.000	116.549	0.300	20.411	0.663	18.153	0.663	3.626	2.478	4.288	1.084	1.542
1.000	48.92	1.000	1.000	0.000	126.201	0.303	20.123	0.699	17.681	0.699	3.810	2.699	4.509	1.173	1.789
1.050	55.01	1.050	1.050	0.000	136.023	0.307	19.826	0.735	17.224	0.735	3.998	2.923	4.733	1.264	2.054
1.100	61.57	1.100	1.100	0.000	146.206	0.310	19.525	0.771	16.758	0.771	4.195	3.156	4.966	1.358	2.347
1.150	68.61	1.150	1.150	0.000	157.357	0.303	19.214	0.807	16.236	0.807	4.456	3.453	5.263	1.462	2.692
1.200	76.21	1.200	1.200	0.000	169.753	0.293	18.887	0.844	15.645	0.844	4.803	3.837	5.647	1.578	3.106
1.250	84.39	1.250	1.250	0.000	182.071	0.297	18.545	0.881	15.071	0.881	5.051	4.122	5.932	1.692	3.562
1.300	93.13	1.300	1.300	0.000	194.135	0.301	18.192	0.918	14.516	0.918	5.199	4.307	6.117	1.801	4.060
1.350	102.2	1.350	1.350	0.000	199.368	0.314	17.865	0.954	14.544	0.954	4.966	4.111	5.921	1.833	4.162
1.400	111.5	1.400	1.400	0.000	204.095	0.329	17.592	0.989	14.605	0.989	4.716	3.895	5.705	1.859	4.238
1.450	120.8	1.450	1.450	0.000	208.713	0.343	17.363	1.023	14.666	1.023	4.487	3.700	5.510	1.883	4.309
1.500	130.3	1.500	1.500	0.000	213.253	0.354	17.170	1.056	14.726	1.056	4.279	3.525	5.335	1.905	4.377
1.550	139.9	1.550	1.550	0.000	217.735	0.364	17.004	1.088	14.786	1.088	4.089	3.367	5.177	1.926	4.443
1.600	149.5	1.600	1.600	0.000	222.163	0.375	16.863	1.119	14.846	1.119	3.915	3.224	5.034	1.945	4.504
1.650	159.3	1.650	1.650	0.000	226.547	0.386	16.741	1.150	14.906	1.150	3.754	3.094	4.904	1.964	4.564
1.700	169.2	1.700	1.700	0.000	230.895	0.396	16.636	1.181	14.965	1.181	3.606	2.977	4.787	1.981	4.621
1.750	179.1	1.750	1.750	0.000	235.213	0.406	16.545	1.211	15.024	1.211	3.469	2.870	4.680	1.997	4.676
1.800	189.1	1.800	1.800	0.000	239.505	0.415	16.465	1.241	15.081	1.241	3.342	2.773	4.583	2.013	4.729
1.850	199.3	1.850	1.850	0.000	243.780	0.425	16.397	1.271	15.138	1.271	3.224	2.685	4.495	2.028	4.781
1.900	209.4	1.900	1.900	0.000	248.053	0.434	16.337	1.300	15.195	1.300	3.116	2.606	4.416	2.043	4.832
1.950	219.7	1.950	1.950	0.000	252.364	0.443	16.285	1.329	15.254	1.329	3.019	2.539	4.349	2.058	4.882
2.000	230.0	2.000	2.000	0.000	256.690	0.451	16.240	1.358	15.313	1.358	2.931	2.479	4.289	2.073	4.931

Fixed Trim = 0.25 m (+ve by stern)

Draft Amidships m	Displacement t	Draft at FP m	Draft at AP m	Trim (+ve by stern) m	Wetted Area m^2	Block coeff. (Cb)	LCB from zero pt. (+ve fwd) m	VCB m	LCF from zero pt. (+ve fwd) m	KB m	BMt m	GMt m	KMt m	Immersion (TPc) tonne/cm	MTc tonne.m
0.100	0.0995	-0.025	0.225	0.250	3.456	0.137	22.032	0.050	22.432	0.050	0.652	-1.080	0.701	0.033	0.006
0.150	0.3538	0.025	0.275	0.250	7.337	0.167	22.429	0.083	22.632	0.083	0.955	-0.741	1.038	0.070	0.019
0.200	0.8078	0.075	0.325	0.250	11.850	0.183	22.527	0.117	22.556	0.117	1.236	-0.424	1.354	0.113	0.038
0.250	1.488	0.125	0.375	0.250	16.876	0.196	22.497	0.153	22.361	0.153	1.501	-0.125	1.654	0.160	0.065
0.300	2.418	0.175	0.425	0.250	22.352	0.205	22.393	0.189	22.095	0.189	1.759	0.169	1.948	0.212	0.100
0.350	3.615	0.225	0.475	0.250	28.132	0.214	22.247	0.225	21.819	0.225	2.004	0.449	2.229	0.267	0.142
0.400	5.090	0.275	0.525	0.250	34.141	0.223	22.081	0.261	21.538	0.261	2.215	0.695	2.476	0.324	0.192
0.450	6.853	0.325	0.575	0.250	40.335	0.233	21.904	0.297	21.248	0.297	2.388	0.903	2.686	0.382	0.250
0.500	8.912	0.375	0.625	0.250	46.725	0.242	21.717	0.333	20.945	0.333	2.535	1.084	2.868	0.442	0.318
0.550	11.27	0.425	0.675	0.250	53.365	0.250	21.521	0.370	20.624	0.370	2.668	1.253	3.038	0.504	0.398
0.600	13.95	0.475	0.725	0.250	60.344	0.257	21.315	0.406	20.277	0.406	2.803	1.422	3.209	0.569	0.492
0.650	16.97	0.525	0.775	0.250	67.750	0.263	21.097	0.442	19.900	0.442	2.946	1.600	3.388	0.638	0.602
0.700	20.35	0.575	0.825	0.250	75.659	0.269	20.864	0.479	19.487	0.479	3.102	1.791	3.581	0.712	0.733
0.750	24.10	0.625	0.875	0.250	84.142	0.275	20.614	0.516	19.036	0.516	3.273	1.998	3.789	0.792	0.889
0.800	28.28	0.675	0.925	0.250	93.265	0.280	20.345	0.553	18.545	0.553	3.462	2.222	4.015	0.878	1.075
0.850	32.89	0.725	0.975	0.250	103.079	0.285	20.055	0.591	18.014	0.591	3.671	2.467	4.262	0.970	1.298
0.900	37.99	0.775	1.025	0.250	113.319	0.290	19.745	0.629	17.482	0.629	3.900	2.732	4.530	1.067	1.547
0.950	43.57	0.825	1.075	0.250	123.788	0.295	19.422	0.668	16.969	0.668	4.139	3.008	4.807	1.165	1.819
1.000	49.64	0.875	1.125	0.250	134.639	0.299	19.090	0.707	16.454	0.707	4.387	3.292	5.094	1.267	2.121
1.050	56.25	0.925	1.175	0.250	146.449	0.290	18.747	0.746	15.887	0.746	4.694	3.635	5.439	1.379	2.479
1.100	63.47	0.975	1.225	0.250	160.204	0.273	18.383	0.785	15.200	0.785	5.173	4.151	5.958	1.510	2.934
1.150	71.36	1.025	1.275	0.250	174.633	0.276	17.991	0.825	14.489	0.825	5.609	4.625	6.434	1.646	3.467
1.200	79.87	1.075	1.325	0.250	185.003	0.283	17.589	0.865	14.143	0.865	5.691	4.744	6.556	1.737	3.826
1.250	88.65	1.125	1.375	0.250	190.326	0.301	17.251	0.903	14.214	0.903	5.397	4.486	6.300	1.772	3.915
1.300	97.59	1.175	1.425	0.250	195.295	0.318	16.977	0.940	14.290	0.940	5.107	4.230	6.047	1.802	3.998
1.350	106.7	1.225	1.475	0.250	200.074	0.332	16.751	0.975	14.364	0.975	4.838	3.995	5.813	1.829	4.076
1.400	115.9	1.275	1.525	0.250	204.741	0.345	16.565	1.008	14.437	1.008	4.595	3.784	5.603	1.854	4.151
1.450	125.2	1.325	1.575	0.250	209.320	0.356	16.409	1.041	14.509	1.041	4.374	3.594	5.415	1.876	4.222
1.500	134.6	1.375	1.625	0.250	213.829	0.368	16.278	1.073	14.579	1.073	4.172	3.423	5.245	1.898	4.290
1.550	144.2	1.425	1.675	0.250	218.283	0.379	16.168	1.105	14.649	1.105	3.987	3.270	5.092	1.918	4.355
1.600	153.8	1.475	1.725	0.250	222.693	0.390	16.075	1.136	14.716	1.136	3.818	3.131	4.954	1.936	4.418
1.650	163.5	1.525	1.775	0.250	227.064	0.400	15.996	1.166	14.783	1.166	3.664	3.006	4.830	1.954	4.478
1.700	173.4	1.575	1.825	0.250	231.403	0.410	15.929	1.196	14.847	1.196	3.521	2.893	4.717	1.971	4.536
1.750	183.2	1.625	1.875	0.250	235.713	0.420	15.873	1.226	14.911	1.226	3.389	2.790	4.615	1.987	4.593
1.800	193.2	1.675	1.925	0.250	240.000	0.430	15.825	1.255	14.973	1.255	3.266	2.697	4.521	2.003	4.647
1.850	203.3	1.725	1.975	0.250	244.268	0.439	15.784	1.284	15.034	1.284	3.152	2.612	4.437	2.017	4.700
1.900	213.4	1.775	2.025	0.250	248.531	0.447	15.750	1.313	15.094	1.313	3.047	2.535	4.360	2.032	4.752
1.950	223.6	1.825	2.075	0.250	252.821	0.456	15.721	1.342	15.156	1.342	2.952	2.469	4.295	2.046	4.803
2.000	233.9	1.875	2.125	0.250	257.139	0.464	15.698	1.371	15.218	1.371	2.867	2.412	4.237	2.061	4.853

Fixed Trim = 0.5 m (+ve by stern)

Draft Amidships m	Displacement t	Draft at FP m	Draft at AP m	Trim (+ve by stern) m	Wetted Area m^2	Block coeff. (Cb)	LCB from zero pt. (+ve fwd) m	VCB m	LCF from zero pt. (+ve fwd) m	KB m	BMt m	GMt m	KMt m	Immersion (TPc) tonne/cm	MTc tonne.m
0.100	0.0374	-0.150	0.350	0.500	1.661	0.164	20.936	0.042	21.068	0.042	0.628	-1.099	0.669	0.016	0.001
0.150	0.1891	-0.100	0.400	0.500	4.733	0.150	21.185	0.075	21.359	0.075	0.998	-0.691	1.073	0.046	0.006
0.200	0.5203	-0.050	0.450	0.500	9.026	0.157	21.368	0.109	21.524	0.109	1.310	-0.343	1.419	0.088	0.020
0.250	1.077	0.000	0.500	0.500	14.088	0.173	21.434	0.144	21.440	0.144	1.592	-0.024	1.737	0.136	0.044
0.300	1.892	0.050	0.550	0.500	19.740	0.184	21.395	0.181	21.251	0.181	1.865	0.284	2.046	0.190	0.078
0.350	2.987	0.100	0.600	0.500	25.800	0.195	21.300	0.218	21.026	0.218	2.128	0.583	2.346	0.248	0.121
0.400	4.379	0.150	0.650	0.500	32.191	0.206	21.171	0.255	20.767	0.255	2.362	0.852	2.617	0.309	0.174
0.450	6.082	0.200	0.700	0.500	38.851	0.217	21.017	0.292	20.476	0.292	2.555	1.080	2.847	0.373	0.239
0.500	8.108	0.250	0.750	0.500	45.794	0.228	20.842	0.330	20.157	0.330	2.718	1.278	3.048	0.438	0.316
0.550	10.47	0.300	0.800	0.500	53.107	0.238	20.647	0.368	19.801	0.368	2.872	1.468	3.240	0.507	0.409
0.600	13.19	0.350	0.850	0.500	60.923	0.246	20.431	0.406	19.398	0.406	3.036	1.667	3.442	0.581	0.522
0.650	16.29	0.400	0.900	0.500	69.385	0.254	20.191	0.444	18.943	0.444	3.222	1.888	3.666	0.662	0.660
0.700	19.82	0.450	0.950	0.500	78.623	0.260	19.924	0.484	18.428	0.484	3.437	2.138	3.920	0.750	0.830
0.750	23.80	0.500	1.000	0.500	88.771	0.267	19.625	0.524	17.850	0.524	3.685	2.423	4.209	0.846	1.041
0.800	28.29	0.550	1.050	0.500	99.649	0.273	19.294	0.564	17.246	0.564	3.972	2.745	4.536	0.950	1.290
0.850	33.31	0.600	1.100	0.500	110.882	0.280	18.941	0.606	16.663	0.606	4.282	3.092	4.888	1.057	1.566
0.900	38.87	0.650	1.150	0.500	122.557	0.282	18.573	0.648	16.083	0.648	4.605	3.451	5.252	1.169	1.878
0.950	45.01	0.700	1.200	0.500	135.289	0.273	18.190	0.690	15.445	0.690	4.988	3.872	5.678	1.290	2.253
1.000	51.82	0.750	1.250	0.500	150.665	0.253	17.778	0.732	14.634	0.732	5.645	4.565	6.376	1.439	2.756
1.050	59.43	0.800	1.300	0.500	167.901	0.253	17.315	0.777	13.725	0.777	6.408	5.366	7.184	1.605	3.399
1.100	67.61	0.850	1.350	0.500	174.834	0.268	16.879	0.820	13.743	0.820	6.187	5.182	7.006	1.660	3.541
1.150	76.01	0.900	1.400	0.500	180.629	0.287	16.537	0.860	13.832	0.860	5.856	4.887	6.716	1.701	3.639
1.200	84.61	0.950	1.450	0.500	186.024	0.305	16.267	0.898	13.925	0.898	5.536	4.601	6.433	1.737	3.730
1.250	93.37	1.000	1.500	0.500	191.098	0.321	16.051	0.934	14.016	0.934	5.232	4.329	6.165	1.768	3.817
1.300	102.3	1.050	1.550	0.500	195.951	0.335	15.878	0.968	14.105	0.968	4.949	4.078	5.917	1.796	3.899
1.350	111.3	1.100	1.600	0.500	200.676	0.348	15.738	1.001	14.191	1.001	4.692	3.853	5.693	1.822	3.978
1.400	120.5	1.150	1.650	0.500	205.304	0.360	15.623	1.034	14.275	1.034	4.459	3.651	5.492	1.845	4.053
1.450	129.8	1.200	1.700	0.500	209.855	0.372	15.530	1.065	14.357	1.065	4.247	3.469	5.312	1.867	4.125
1.500	139.2	1.250	1.750	0.500	214.347	0.383	15.453	1.096	14.436	1.096	4.054	3.306	5.150	1.888	4.195
1.550	148.7	1.300	1.800	0.500	218.790	0.394	15.391	1.127	14.512	1.127	3.879	3.160	5.005	1.908	4.261
1.600	158.2	1.350	1.850	0.500	223.192	0.405	15.340	1.157	14.586	1.157	3.718	3.028	4.874	1.926	4.325
1.650	167.9	1.400	1.900	0.500	227.558	0.415	15.298	1.186	14.658	1.186	3.570	2.910	4.756	1.944	4.387
1.700	177.7	1.450	1.950	0.500	231.893	0.425	15.265	1.216	14.728	1.216	3.433	2.802	4.649	1.960	4.447
1.750	187.5	1.500	2.000	0.500	236.200	0.434	15.239	1.245	14.796	1.245	3.306	2.704	4.551	1.976	4.505
1.800	197.4	1.550	2.050	0.500	240.485	0.444	15.218	1.274	14.863	1.274	3.188	2.614	4.462	1.991	4.561
1.850	207.4	1.600	2.100	0.500	244.752	0.453	15.203	1.302	14.928	1.302	3.079	2.533	4.381	2.006	4.616
1.900	217.5	1.650	2.150	0.500	249.011	0.461	15.192	1.331	14.993	1.331	2.977	2.460	4.308	2.020	4.668
1.950	227.6	1.700	2.200	0.500	253.287	0.469	15.184	1.359	15.057	1.359	2.885	2.396	4.244	2.034	4.720
2.000	237.8	1.750	2.250	0.500	257.596	0.477	15.180	1.387	15.122	1.387	2.802	2.341	4.189	2.048	4.771

Fixed Trim = 0.75 m (+ve by stern)

Draft Amidships m	Displacement T	Draft at FP m	Draft at AP m	Trim (+ve by stern) m	Wetted Area m^2	Block coeff. (Cb)	LCB from zero pt. (+ve fwd) m	VCB m	LCF from zero pt. (+ve fwd) m	KB m	BMt m	GMt m	KMt m	Immersion (TPc) tonne/cm	MTc tonne.m
0.100	0.0135	-0.275	0.475	0.750	0.826	0.193	20.536	0.034	20.555	0.034	0.521	-1.201	0.555	0.008	0.000
0.150	0.1084	-0.225	0.525	0.750	3.215	0.177	20.570	0.070	20.568	0.070	0.960	-0.724	1.030	0.032	0.002
0.200	0.3550	-0.175	0.575	0.750	6.940	0.171	20.551	0.107	20.507	0.107	1.341	-0.307	1.448	0.069	0.009
0.250	0.8138	-0.125	0.625	0.750	11.858	0.168	20.495	0.145	20.392	0.145	1.674	0.064	1.819	0.117	0.027
0.300	1.534	-0.075	0.675	0.750	17.609	0.178	20.420	0.183	20.271	0.183	1.979	0.404	2.162	0.172	0.058
0.350	2.546	-0.025	0.725	0.750	23.927	0.188	20.324	0.222	20.078	0.222	2.266	0.728	2.487	0.234	0.101
0.400	3.876	0.025	0.775	0.750	30.751	0.197	20.198	0.260	19.833	0.260	2.526	1.023	2.785	0.299	0.158
0.450	5.545	0.075	0.825	0.750	38.010	0.208	20.044	0.299	19.537	0.299	2.745	1.278	3.044	0.369	0.231
0.500	7.572	0.125	0.875	0.750	45.727	0.220	19.861	0.338	19.182	0.338	2.939	1.508	3.277	0.443	0.322
0.550	9.983	0.175	0.925	0.750	54.064	0.230	19.646	0.378	18.757	0.378	3.137	1.742	3.515	0.523	0.438
0.600	12.81	0.225	0.975	0.750	63.254	0.239	19.394	0.419	18.247	0.419	3.367	2.007	3.786	0.611	0.585
0.650	16.11	0.275	1.025	0.750	73.560	0.247	19.097	0.462	17.640	0.462	3.648	2.325	4.109	0.710	0.776
0.700	19.94	0.325	1.075	0.750	85.039	0.255	18.751	0.505	16.957	0.505	3.998	2.711	4.503	0.821	1.018
0.750	24.33	0.375	1.125	0.750	97.148	0.263	18.365	0.550	16.289	0.550	4.407	3.157	4.956	0.938	1.296
0.800	29.33	0.425	1.175	0.750	109.854	0.266	17.954	0.596	15.625	0.596	4.845	3.632	5.440	1.061	1.616
0.850	34.96	0.475	1.225	0.750	123.892	0.254	17.520	0.642	14.888	0.642	5.367	4.190	6.007	1.197	2.013
0.900	41.36	0.525	1.275	0.750	141.730	0.234	17.039	0.689	13.880	0.689	6.340	5.201	7.028	1.371	2.590
0.950	48.68	0.575	1.325	0.750	157.386	0.232	16.485	0.739	13.180	0.739	7.167	6.066	7.904	1.521	3.104
1.000	56.42	0.625	1.375	0.750	164.095	0.253	16.038	0.785	13.275	0.785	6.756	5.692	7.540	1.574	3.223
1.050	64.40	0.675	1.425	0.750	170.282	0.274	15.702	0.827	13.386	0.827	6.357	5.327	7.182	1.620	3.331
1.100	72.61	0.725	1.475	0.750	176.091	0.293	15.447	0.865	13.500	0.865	5.989	4.993	6.853	1.662	3.432
1.150	81.01	0.775	1.525	0.750	181.575	0.311	15.251	0.902	13.613	0.902	5.651	4.686	6.551	1.699	3.528
1.200	89.58	0.825	1.575	0.750	186.770	0.326	15.099	0.937	13.722	0.937	5.336	4.404	6.272	1.732	3.620
1.250	98.31	0.875	1.625	0.750	191.722	0.340	14.982	0.970	13.826	0.970	5.044	4.142	6.013	1.761	3.707
1.300	107.2	0.925	1.675	0.750	196.513	0.353	14.890	1.002	13.926	1.002	4.776	3.905	5.777	1.787	3.791
1.350	116.2	0.975	1.725	0.750	201.200	0.366	14.819	1.034	14.023	1.034	4.534	3.692	5.566	1.812	3.871
1.400	125.3	1.025	1.775	0.750	205.805	0.378	14.765	1.064	14.116	1.064	4.314	3.502	5.377	1.835	3.948
1.450	134.5	1.075	1.825	0.750	210.343	0.389	14.724	1.095	14.205	1.095	4.115	3.332	5.208	1.857	4.022
1.500	143.9	1.125	1.875	0.750	214.828	0.401	14.693	1.124	14.291	1.124	3.933	3.180	5.056	1.877	4.093
1.550	153.3	1.175	1.925	0.750	219.267	0.411	14.671	1.153	14.374	1.153	3.767	3.042	4.920	1.896	4.162
1.600	162.8	1.225	1.975	0.750	223.667	0.421	14.656	1.182	14.454	1.182	3.614	2.918	4.796	1.915	4.228
1.650	172.4	1.275	2.025	0.750	228.033	0.431	14.647	1.211	14.532	1.211	3.474	2.806	4.684	1.932	4.292
1.700	182.1	1.325	2.075	0.750	232.368	0.441	14.642	1.240	14.607	1.240	3.343	2.704	4.582	1.948	4.354
1.750	191.9	1.375	2.125	0.750	236.677	0.450	14.643	1.268	14.681	1.268	3.222	2.611	4.489	1.964	4.413
1.800	201.8	1.425	2.175	0.750	240.964	0.459	14.646	1.296	14.752	1.296	3.109	2.527	4.405	1.979	4.471
1.850	211.7	1.475	2.225	0.750	245.232	0.467	14.653	1.324	14.821	1.324	3.004	2.450	4.328	1.994	4.527
1.900	221.7	1.525	2.275	0.750	249.491	0.475	14.662	1.352	14.889	1.352	2.907	2.380	4.258	2.007	4.581
1.950	231.8	1.575	2.325	0.750	253.760	0.483	14.673	1.379	14.956	1.379	2.818	2.319	4.197	2.021	4.635
2.000	241.9	1.625	2.375	0.750	258.059	0.491	14.686	1.407	15.024	1.407	2.738	2.267	4.144	2.036	4.687

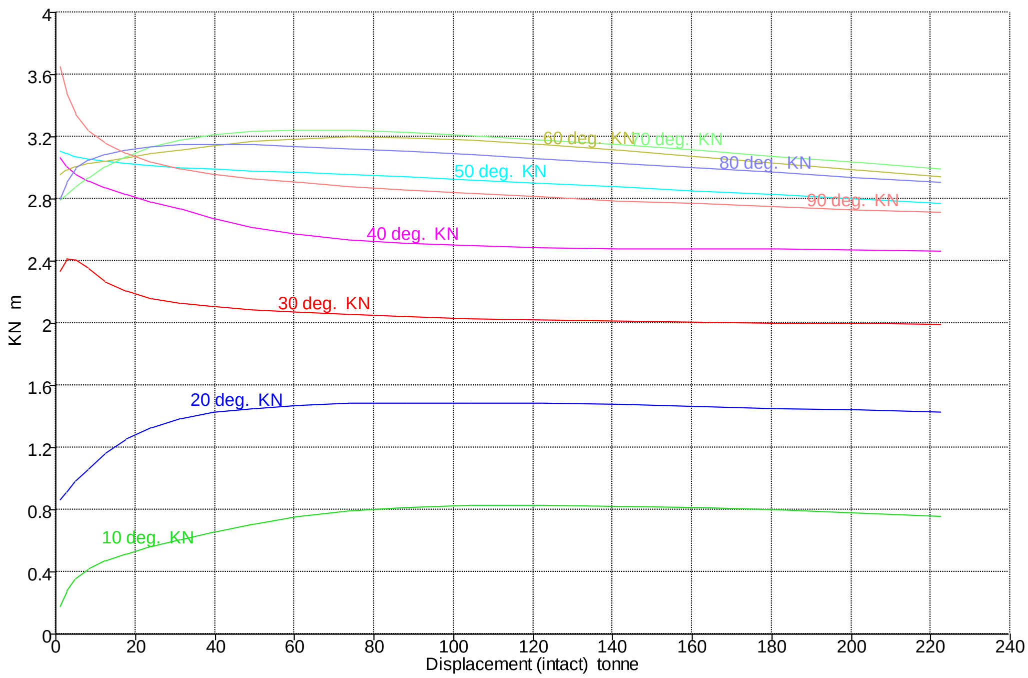
Fixed Trim = 1 m (+ve by stern)

Draft Amidships m	Displacement t	Draft at FP m	Draft at AP m	Trim (+ve by stern) m	Wetted Area m^2	Block coeff. (Cb)	LCB from zero pt. (+ve fwd) m	VCB m	LCF from zero pt. (+ve fwd) m	KB m	BMt m	GMt m	KMt m	Immersion (TPc) tonne/cm	MTc tonne.m
0.100	0.0036	-0.400	0.600	1.000	0.347	0.221	20.361	0.025	20.332	0.025	0.377	-1.340	0.402	0.003	0.000
0.150	0.0630	-0.350	0.650	1.000	2.291	0.199	20.203	0.067	20.099	0.067	0.878	-0.802	0.944	0.023	0.001
0.200	0.2585	-0.300	0.700	1.000	5.769	0.194	19.998	0.109	19.794	0.109	1.326	-0.316	1.435	0.058	0.006
0.250	0.6611	-0.250	0.750	1.000	10.558	0.195	19.765	0.153	19.476	0.153	1.724	0.119	1.876	0.105	0.019
0.300	1.323	-0.200	0.800	1.000	16.273	0.195	19.558	0.195	19.238	0.195	2.082	0.512	2.276	0.161	0.045
0.350	2.288	-0.150	0.850	1.000	22.875	0.193	19.373	0.237	19.006	0.237	2.407	0.874	2.642	0.226	0.087
0.400	3.595	-0.100	0.900	1.000	30.251	0.201	19.191	0.278	18.730	0.278	2.701	1.205	2.978	0.298	0.150
0.450	5.277	-0.050	0.950	1.000	38.329	0.210	18.988	0.320	18.373	0.320	2.967	1.506	3.285	0.377	0.235
0.500	7.372	0.000	1.000	1.000	47.288	0.219	18.749	0.362	17.917	0.362	3.230	1.805	3.591	0.464	0.352
0.550	9.931	0.050	1.050	1.000	57.507	0.227	18.460	0.407	17.326	0.407	3.540	2.152	3.946	0.563	0.512
0.600	13.03	0.100	1.100	1.000	69.399	0.237	18.101	0.454	16.578	0.454	3.950	2.599	4.402	0.679	0.735
0.650	16.74	0.150	1.150	1.000	82.433	0.247	17.675	0.503	15.809	0.503	4.482	3.167	4.983	0.807	1.008
0.700	21.11	0.200	1.200	1.000	96.382	0.251	17.208	0.553	15.043	0.553	5.091	3.814	5.642	0.943	1.334
0.750	26.20	0.250	1.250	1.000	112.249	0.234	16.704	0.605	14.162	0.605	5.849	4.610	6.451	1.099	1.760
0.800	32.20	0.300	1.300	1.000	134.963	0.214	16.109	0.659	12.755	0.659	7.705	6.504	8.362	1.324	2.487
0.850	39.13	0.350	1.350	1.000	145.873	0.217	15.484	0.715	12.597	0.715	8.011	6.849	8.724	1.423	2.741
0.900	46.39	0.400	1.400	1.000	152.937	0.241	15.042	0.763	12.726	0.763	7.447	6.320	8.207	1.481	2.870
0.950	53.92	0.450	1.450	1.000	159.454	0.263	14.728	0.806	12.865	0.806	6.935	5.842	7.738	1.532	2.989
1.000	61.69	0.500	1.500	1.000	165.593	0.283	14.502	0.845	13.004	0.845	6.486	5.425	7.328	1.577	3.101
1.050	69.68	0.550	1.550	1.000	171.428	0.302	14.338	0.881	13.142	0.881	6.090	5.061	6.969	1.619	3.208
1.100	77.87	0.600	1.600	1.000	176.993	0.319	14.220	0.915	13.275	0.915	5.737	4.739	6.650	1.657	3.309
1.150	86.24	0.650	1.650	1.000	182.305	0.334	14.134	0.948	13.403	0.948	5.416	4.448	6.362	1.692	3.407
1.200	94.78	0.700	1.700	1.000	187.379	0.348	14.074	0.980	13.524	0.980	5.119	4.182	6.097	1.723	3.500
1.250	103.5	0.750	1.750	1.000	192.255	0.361	14.033	1.011	13.640	1.011	4.845	3.937	5.854	1.751	3.589
1.300	112.3	0.800	1.800	1.000	197.009	0.374	14.006	1.041	13.750	1.041	4.596	3.718	5.635	1.776	3.674
1.350	121.2	0.850	1.850	1.000	201.674	0.386	13.991	1.070	13.855	1.070	4.371	3.522	5.440	1.801	3.757
1.400	130.3	0.900	1.900	1.000	206.267	0.398	13.985	1.099	13.955	1.099	4.167	3.346	5.264	1.823	3.836
1.450	139.4	0.950	1.950	1.000	210.800	0.409	13.986	1.128	14.052	1.128	3.980	3.188	5.107	1.844	3.912
1.500	148.7	1.000	2.000	1.000	215.282	0.419	13.993	1.156	14.144	1.156	3.810	3.046	4.965	1.865	3.986
1.550	158.1	1.050	2.050	1.000	219.722	0.429	14.005	1.184	14.234	1.184	3.654	2.918	4.836	1.884	4.057
1.600	167.5	1.100	2.100	1.000	224.124	0.439	14.020	1.212	14.320	1.212	3.509	2.802	4.720	1.902	4.125
1.650	177.1	1.150	2.150	1.000	228.493	0.449	14.039	1.240	14.404	1.240	3.376	2.697	4.614	1.919	4.191
1.700	186.7	1.200	2.200	1.000	232.833	0.458	14.060	1.267	14.484	1.267	3.252	2.602	4.518	1.936	4.255
1.750	196.4	1.250	2.250	1.000	237.146	0.466	14.083	1.294	14.562	1.294	3.137	2.514	4.430	1.951	4.317
1.800	206.2	1.300	2.300	1.000	241.437	0.475	14.107	1.322	14.638	1.322	3.030	2.435	4.350	1.966	4.377
1.850	216.1	1.350	2.350	1.000	245.710	0.483	14.133	1.349	14.711	1.349	2.930	2.363	4.277	1.981	4.435
1.900	226.0	1.400	2.400	1.000	249.972	0.490	14.160	1.376	14.783	1.376	2.837	2.297	4.211	1.994	4.491
1.950	236.0	1.450	2.450	1.000	254.238	0.498	14.188	1.403	14.854	1.403	2.751	2.240	4.153	2.008	4.546
2.000	246.1	1.500	2.500	1.000	258.527	0.505	14.217	1.430	14.924	1.430	2.674	2.190	4.102	2.022	4.600

KN calculation

Fixed Trim =-0.5 m (+ve by stern)

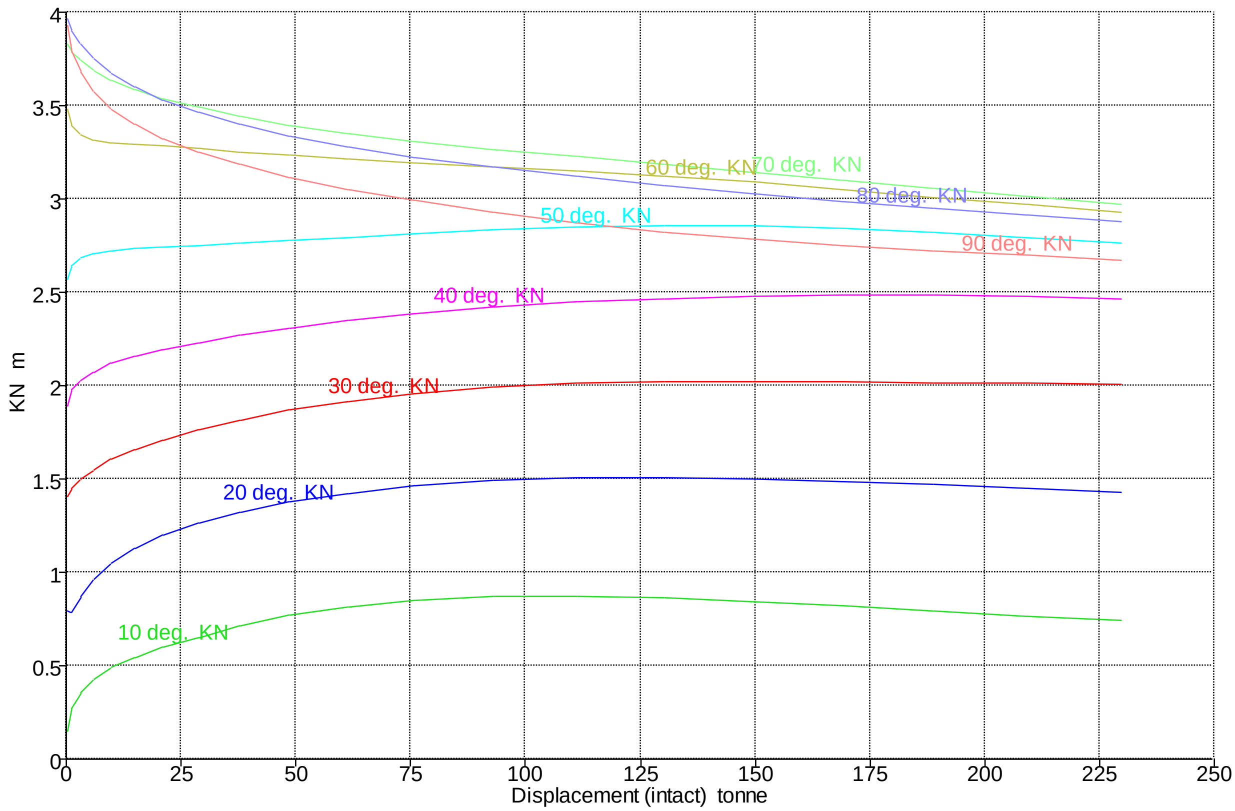
Specific gravity = 1.025; (Density = 1.025 tonne/m^3)
VCG = 0 m; TCG = 0 m



Displacement (intact) tonne	Draft Amidships m	LCG m	TCG m	Assumed VCG m	KN 10.0 deg. Starb.	KN 20.0 deg. Starb.	KN 30.0 deg. Starb.	KN 40.0 deg. Starb.	KN 50.0 deg. Starb.	KN 60.0 deg. Starb.	KN 70.0 deg. Starb.	KN 80.0 deg. Starb.	KN 90.0 deg. Starb.
1.101	0.100	26.084	0.000	0.000	0.182	0.863	2.333	3.065	3.106	2.958	2.796	2.800	3.651
2.670	0.200	25.291	0.000	0.000	0.281	0.911	2.411	3.008	3.090	2.985	2.829	2.909	3.476
5.071	0.300	24.709	0.000	0.000	0.356	0.983	2.411	2.957	3.074	3.007	2.878	2.999	3.342
8.385	0.400	24.209	0.000	0.000	0.418	1.064	2.352	2.911	3.057	3.026	2.939	3.050	3.235
12.65	0.500	23.760	0.000	0.000	0.470	1.162	2.265	2.869	3.041	3.046	3.004	3.087	3.156
17.89	0.600	23.342	0.000	0.000	0.517	1.257	2.204	2.826	3.025	3.067	3.071	3.114	3.090
24.17	0.700	22.939	0.000	0.000	0.563	1.331	2.161	2.782	3.012	3.091	3.133	3.134	3.035
31.56	0.800	22.536	0.000	0.000	0.609	1.386	2.129	2.734	3.000	3.116	3.180	3.147	2.993
40.14	0.900	22.120	0.000	0.000	0.659	1.425	2.105	2.674	2.990	3.143	3.213	3.152	2.959
50.03	1.000	21.683	0.000	0.000	0.711	1.453	2.085	2.617	2.981	3.169	3.234	3.148	2.930
61.32	1.100	21.220	0.000	0.000	0.755	1.471	2.069	2.572	2.972	3.188	3.243	3.139	2.904
74.13	1.200	20.730	0.000	0.000	0.790	1.483	2.055	2.539	2.960	3.196	3.241	3.124	2.879
88.49	1.300	20.224	0.000	0.000	0.813	1.488	2.042	2.515	2.942	3.194	3.229	3.105	2.856
104.6	1.400	19.692	0.000	0.000	0.826	1.488	2.032	2.498	2.921	3.179	3.209	3.082	2.833
122.3	1.500	19.146	0.000	0.000	0.828	1.483	2.022	2.487	2.900	3.152	3.182	3.057	2.811
141.5	1.600	18.614	0.000	0.000	0.823	1.475	2.014	2.481	2.877	3.115	3.149	3.028	2.789
161.3	1.700	18.194	0.000	0.000	0.812	1.465	2.007	2.478	2.853	3.072	3.113	2.999	2.769
181.4	1.800	17.870	0.000	0.000	0.798	1.453	2.002	2.477	2.827	3.027	3.074	2.969	2.749
201.9	1.900	17.617	0.000	0.000	0.780	1.441	1.996	2.471	2.801	2.984	3.033	2.939	2.730
222.7	2.000	17.417	0.000	0.000	0.760	1.428	1.992	2.461	2.773	2.944	2.991	2.908	2.712

Initial Trim = 0 m (+ve by stern)

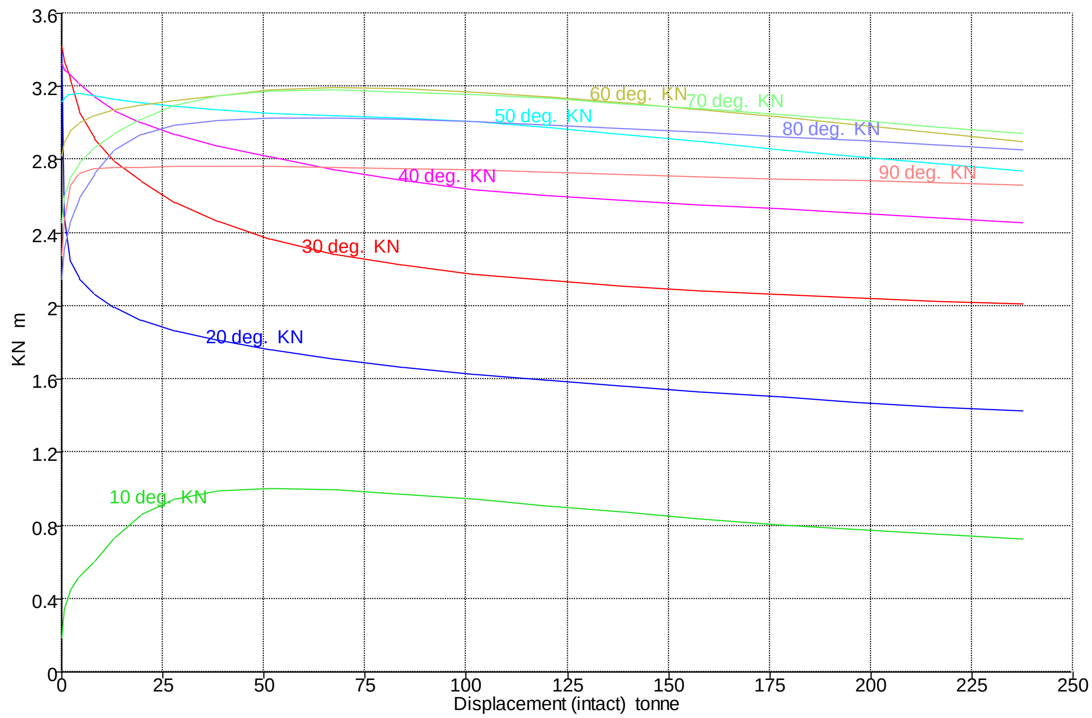
Specific gravity = 1.025; (Density = 1.025 tonne/m^3)
VCG = 0 m; TCG = 0 m



Displacement (intact) tonne	Draft Amidships m	KN 10.0 deg. Starb.	KN 20.0 deg. Starb.	KN 30.0 deg. Starb.	KN 40.0 deg. Starb.	KN 50.0 deg. Starb.	KN 60.0 deg. Starb.	KN 70.0 deg. Starb.	KN 80.0 deg. Starb.	KN 90.0 deg. Starb.
0.2687	0.100	0.151	0.796	1.407	1.894	2.571	3.476	3.827	3.968	3.926
1.254	0.200	0.271	0.784	1.453	1.979	2.640	3.393	3.785	3.901	3.796
3.122	0.300	0.360	0.870	1.497	2.028	2.684	3.344	3.743	3.827	3.679
5.999	0.400	0.432	0.963	1.552	2.072	2.705	3.317	3.689	3.749	3.571
9.949	0.500	0.491	1.049	1.607	2.119	2.720	3.300	3.635	3.673	3.479
15.02	0.600	0.545	1.130	1.659	2.158	2.732	3.292	3.585	3.599	3.399
21.31	0.700	0.598	1.202	1.711	2.194	2.743	3.284	3.538	3.529	3.324
28.94	0.800	0.653	1.267	1.764	2.230	2.753	3.271	3.490	3.462	3.252
38.08	0.900	0.714	1.325	1.818	2.268	2.764	3.253	3.442	3.397	3.182
48.92	1.000	0.770	1.378	1.869	2.308	2.778	3.233	3.394	3.335	3.115
61.57	1.100	0.815	1.424	1.915	2.348	2.795	3.211	3.349	3.277	3.051
76.21	1.200	0.850	1.462	1.955	2.386	2.816	3.190	3.307	3.224	2.989
93.13	1.300	0.871	1.493	1.990	2.421	2.837	3.170	3.266	3.171	2.926
111.5	1.400	0.875	1.508	2.012	2.448	2.852	3.149	3.227	3.119	2.870
130.3	1.500	0.864	1.508	2.021	2.465	2.858	3.122	3.186	3.072	2.824
149.5	1.600	0.844	1.499	2.024	2.476	2.854	3.089	3.144	3.029	2.786
169.2	1.700	0.820	1.485	2.022	2.483	2.841	3.051	3.101	2.989	2.753
189.1	1.800	0.793	1.468	2.017	2.485	2.820	3.011	3.057	2.951	2.723
209.4	1.900	0.767	1.450	2.011	2.478	2.793	2.969	3.013	2.914	2.697
230.0	2.000	0.744	1.431	2.004	2.463	2.762	2.926	2.969	2.879	2.673

Fixed Trim = 0.5 m (+ve by stern)

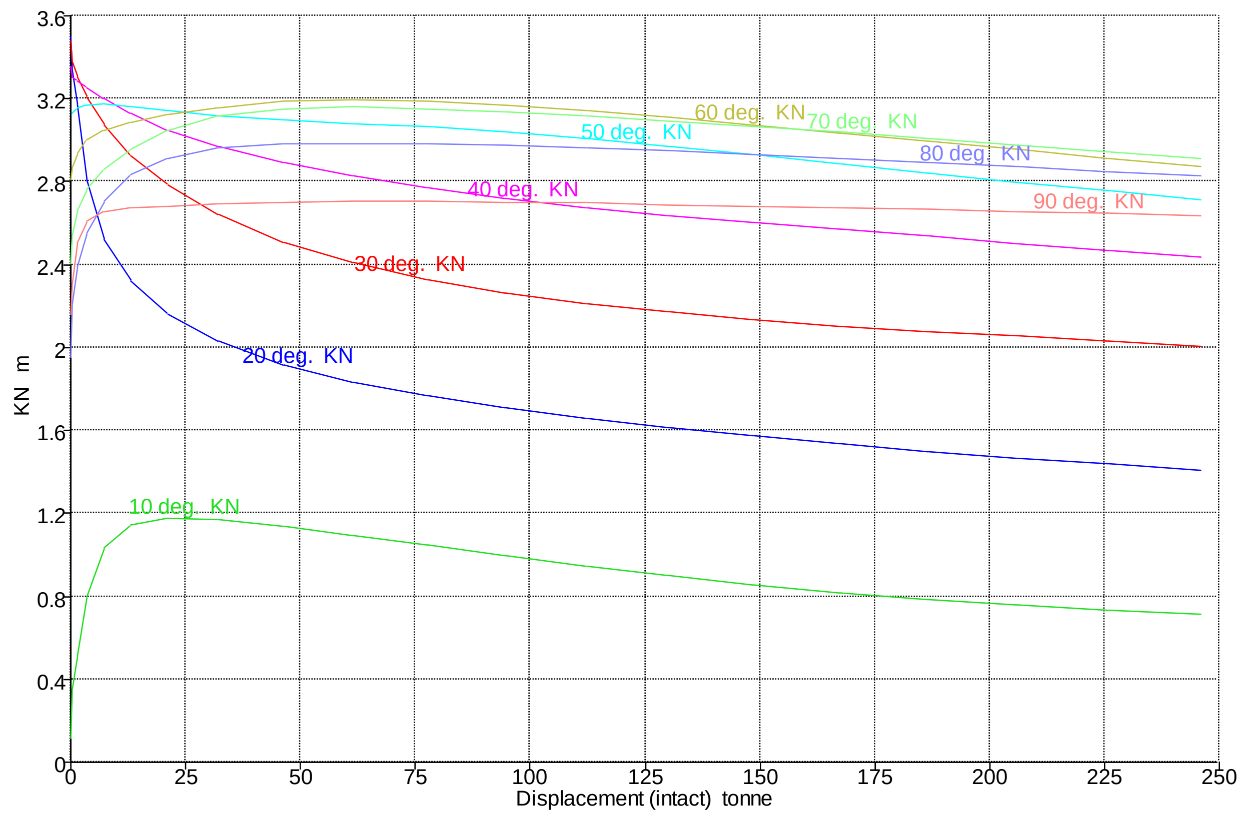
Specific gravity = 1.025; (Density = 1.025 tonne/m^3)
VCG = 0 m; TCG = 0 m



Displacement (intact) tonne	Draft Amidships m	LCG m	TCG m	Assumed VCG m	KN 10.0 deg. Starb.	KN 20.0 deg. Starb.	KN 30.0 deg. Starb.	KN 40.0 deg. Starb.	KN 50.0 deg. Starb.	KN 60.0 deg. Starb.	KN 70.0 deg. Starb.	KN 80.0 deg. Starb.	KN 90.0 deg. Starb.
0.0374	0.100	20.935	0.000	0.000	0.187	3.395	3.419	3.321	3.116	2.819	2.458	2.144	2.276
0.5203	0.200	21.366	0.000	0.000	0.343	2.494	3.343	3.290	3.134	2.894	2.597	2.306	2.464
1.892	0.300	21.392	0.000	0.000	0.445	2.251	3.243	3.266	3.156	2.955	2.697	2.453	2.652
4.379	0.400	21.168	0.000	0.000	0.523	2.147	3.062	3.215	3.162	3.004	2.785	2.589	2.723
8.108	0.500	20.837	0.000	0.000	0.601	2.065	2.914	3.142	3.151	3.042	2.867	2.723	2.749
13.19	0.600	20.426	0.000	0.000	0.736	1.991	2.791	3.069	3.132	3.072	2.945	2.853	2.757
19.82	0.700	19.917	0.000	0.000	0.865	1.925	2.678	3.002	3.112	3.098	3.022	2.938	2.760
28.29	0.800	19.287	0.000	0.000	0.946	1.866	2.567	2.939	3.092	3.124	3.098	2.988	2.763
38.87	0.900	18.564	0.000	0.000	0.989	1.811	2.460	2.876	3.073	3.153	3.151	3.014	2.765
51.82	1.000	17.768	0.000	0.000	1.003	1.760	2.365	2.813	3.057	3.183	3.177	3.026	2.765
67.61	1.100	16.868	0.000	0.000	0.995	1.711	2.285	2.747	3.042	3.196	3.181	3.028	2.760
84.61	1.200	16.254	0.000	0.000	0.972	1.667	2.223	2.687	3.030	3.189	3.172	3.021	2.752
102.3	1.300	15.865	0.000	0.000	0.942	1.629	2.176	2.639	3.010	3.169	3.156	3.007	2.742
120.5	1.400	15.609	0.000	0.000	0.909	1.593	2.139	2.602	2.980	3.142	3.134	2.990	2.732
139.2	1.500	15.438	0.000	0.000	0.874	1.561	2.108	2.576	2.940	3.109	3.108	2.971	2.720
158.2	1.600	15.324	0.000	0.000	0.838	1.530	2.083	2.554	2.897	3.072	3.079	2.950	2.708
177.7	1.700	15.248	0.000	0.000	0.804	1.501	2.062	2.532	2.855	3.031	3.047	2.928	2.696
197.4	1.800	15.201	0.000	0.000	0.775	1.474	2.043	2.508	2.816	2.988	3.014	2.904	2.684
217.5	1.900	15.173	0.000	0.000	0.750	1.449	2.027	2.483	2.778	2.944	2.979	2.880	2.672
237.8	2.000	15.161	0.000	0.000	0.729	1.424	2.012	2.456	2.741	2.902	2.942	2.855	2.660

Fixed Trim = 1 m (+ve by stern)

Specific gravity = 1.025; (Density = 1.025 tonne/m^3)
VCG = 0 m; TCG = 0 m



Displacement (intact) tonne	Draft Amidships m	LCG m	TCG m	Assumed VCG m	KN 10.0 deg. Starb.	KN 20.0 deg. Starb.	KN 30.0 deg. Starb.	KN 40.0 deg. Starb.	KN 50.0 deg. Starb.	KN 60.0 deg. Starb.	KN 70.0 deg. Starb.	KN 80.0 deg. Starb.	KN 90.0 deg. Starb.
0.0036	0.100	20.360	0.000	0.000	0.119	3.497	3.481	3.355	3.128	2.807	2.406	1.954	2.160
0.2585	0.200	19.995	0.000	0.000	0.346	3.340	3.382	3.307	3.131	2.868	2.540	2.206	2.304
1.323	0.300	19.553	0.000	0.000	0.516	3.172	3.305	3.285	3.155	2.939	2.664	2.394	2.509
3.595	0.400	19.183	0.000	0.000	0.794	2.812	3.206	3.252	3.172	2.999	2.767	2.554	2.607
7.372	0.500	18.739	0.000	0.000	1.037	2.521	3.072	3.200	3.174	3.048	2.862	2.705	2.652
13.03	0.600	18.088	0.000	0.000	1.142	2.319	2.923	3.129	3.163	3.088	2.954	2.833	2.671
21.11	0.700	17.193	0.000	0.000	1.178	2.162	2.780	3.050	3.142	3.123	3.046	2.914	2.683
32.20	0.800	16.091	0.000	0.000	1.173	2.030	2.641	2.971	3.119	3.157	3.117	2.961	2.692
46.39	0.900	15.020	0.000	0.000	1.140	1.919	2.510	2.895	3.097	3.187	3.152	2.982	2.700
61.69	1.000	14.478	0.000	0.000	1.095	1.835	2.408	2.828	3.080	3.197	3.160	2.986	2.704
77.87	1.100	14.194	0.000	0.000	1.045	1.767	2.328	2.769	3.063	3.189	3.152	2.983	2.705
94.78	1.200	14.047	0.000	0.000	0.995	1.709	2.264	2.718	3.040	3.170	3.137	2.975	2.702
112.3	1.300	13.977	0.000	0.000	0.945	1.658	2.213	2.674	3.009	3.143	3.117	2.963	2.697
130.3	1.400	13.955	0.000	0.000	0.897	1.613	2.171	2.637	2.971	3.110	3.094	2.948	2.690
148.7	1.500	13.961	0.000	0.000	0.853	1.572	2.135	2.602	2.929	3.075	3.068	2.931	2.682
167.5	1.600	13.987	0.000	0.000	0.816	1.535	2.105	2.569	2.884	3.037	3.040	2.912	2.674
186.7	1.700	14.024	0.000	0.000	0.784	1.500	2.079	2.536	2.839	2.998	3.010	2.892	2.665
206.2	1.800	14.070	0.000	0.000	0.757	1.468	2.055	2.504	2.796	2.957	2.978	2.871	2.655
226.0	1.900	14.122	0.000	0.000	0.734	1.438	2.032	2.471	2.755	2.915	2.945	2.849	2.645
246.1	2.000	14.177	0.000	0.000	0.715	1.410	2.008	2.437	2.716	2.873	2.911	2.827	2.635

Limiting KG

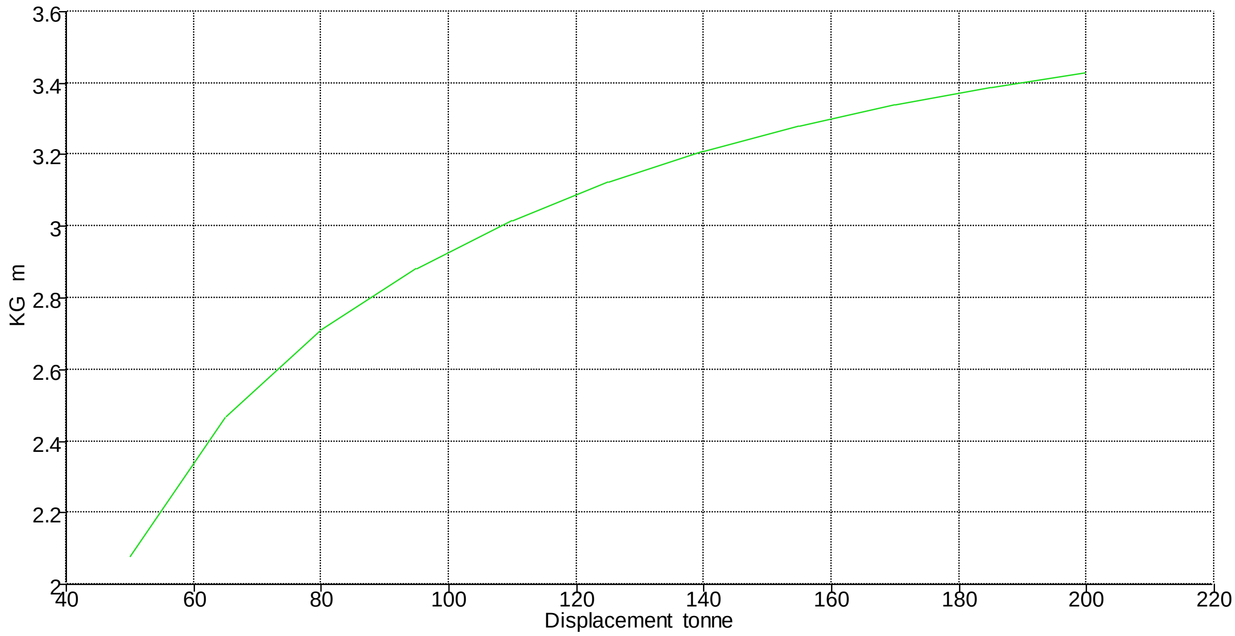
Fixed Trim =-0.5 m (+ve by stern)

Specific gravity = 1.025; (Density = 1.025 tonne/m^3)
Heel to starboard; heel range: from -50 deg to 90 deg in steps of 10 deg.

Criteria tested:

267(85) Ch2 - General Criteria	2.2.1: Area 0 to 30
267(85) Ch2 - General Criteria	2.2.1: Area 0 to 40
267(85) Ch2 - General Criteria	2.2.1: Area 30 to 40
267(85) Ch2 - General Criteria	2.2.2: Max GZ at 30 or greater
267(85) Ch2 - General Criteria	2.2.3: Angle of maximum GZ
267(85) Ch2 - General Criteria	2.2.4: Initial GMt
267(85) Ch2 - General Criteria	2.3: Severe wind and rolling

Displacement (intact) tonne	Draft Amidships m	LCG m	TCG m	VCG m	Limit KG m	min. GM m	Criterion	Name
50.00	1.000	21.655	0.000	2.078	2.078	1.785	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
65.00	1.130	21.041	0.000	2.467	2.467	1.765	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
80.00	1.242	20.481	0.000	2.709	2.709	1.854	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
95.00	1.342	19.965	0.000	2.882	2.882	2.071	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
110.0	1.432	19.478	0.000	3.016	3.016	2.156	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
125.0	1.515	19.023	0.000	3.123	3.123	2.153	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
140.0	1.592	18.606	0.000	3.209	3.209	1.988	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
155.0	1.669	18.268	0.000	3.281	3.281	1.715	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
170.0	1.744	17.996	0.000	3.340	3.340	1.485	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
185.0	1.818	17.773	0.000	3.389	3.389	1.287	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
200.0	1.891	17.589	0.000	3.428	3.428	1.121	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling



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KG m
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200
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Displacement tonne

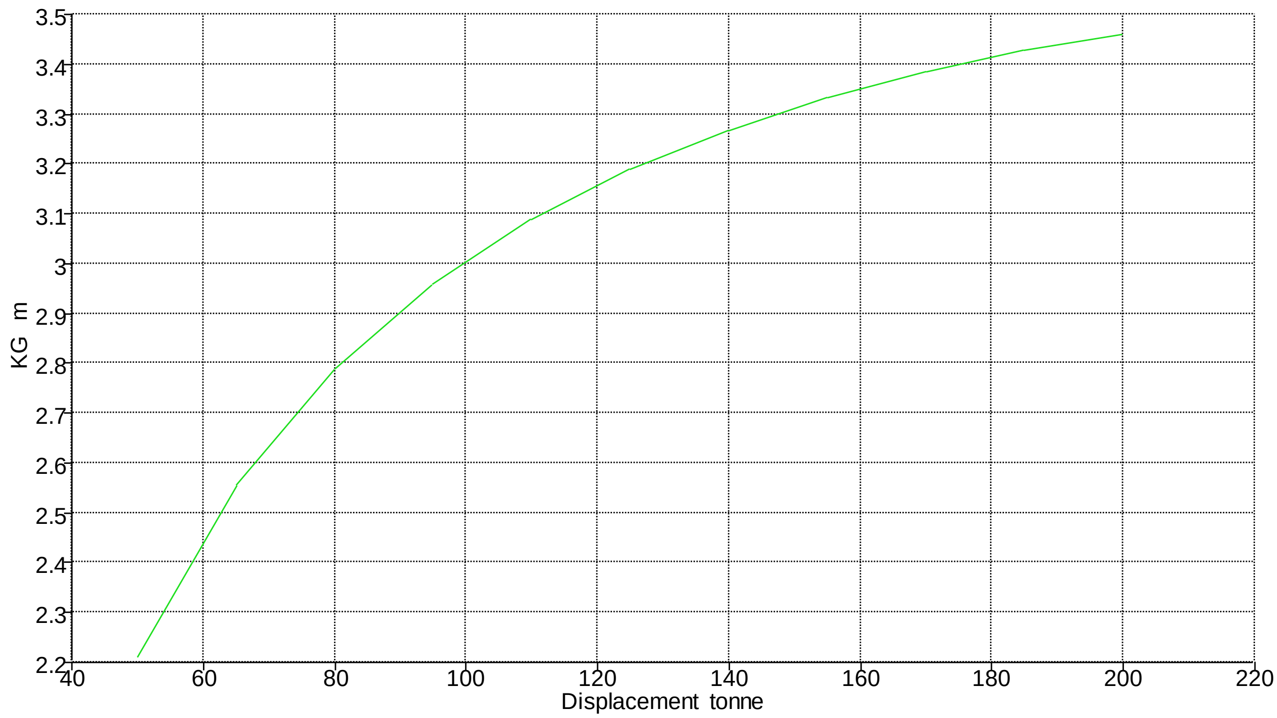
Fixed Trim = 0 m (+ve by stern)

Specific gravity = 1.025; (Density = 1.025 tonne/m^3)
Heel to starboard; heel range: from -50 deg to 90 deg in steps of 10 deg.

Criteria tested:

267(85) Ch2 - General Criteria	2.2.1: Area 0 to 30
267(85) Ch2 - General Criteria	2.2.1: Area 0 to 40
267(85) Ch2 - General Criteria	2.2.1: Area 30 to 40
267(85) Ch2 - General Criteria	2.2.2: Max GZ at 30 or greater
267(85) Ch2 - General Criteria	2.2.3: Angle of maximum GZ
267(85) Ch2 - General Criteria	2.2.4: Initial GMt
267(85) Ch2 - General Criteria	2.3: Severe wind and rolling

Displacement (intact) tonne	Draft Amidships m	LCG m	TCG m	VCG m	Limit KG m	min. GM m	Criterion	Name
50.00	1.009	20.069	0.000	2.211	2.211	2.339	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
65.00	1.125	19.372	0.000	2.557	2.557	2.541	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
80.00	1.224	18.727	0.000	2.789	2.789	3.006	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
95.00	1.310	18.119	0.000	2.959	2.959	3.141	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
110.0	1.392	17.632	0.000	3.089	3.089	2.649	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
125.0	1.472	17.273	0.000	3.189	3.189	2.241	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
140.0	1.551	17.002	0.000	3.268	3.268	1.907	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
155.0	1.628	16.792	0.000	3.332	3.332	1.627	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
170.0	1.704	16.628	0.000	3.385	3.385	1.392	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
185.0	1.779	16.497	0.000	3.427	3.427	1.195	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
200.0	1.854	16.392	0.000	3.461	3.461	1.028	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling



KG m
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2.3
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60
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120
140
160
180
200
220
Displacement tonne

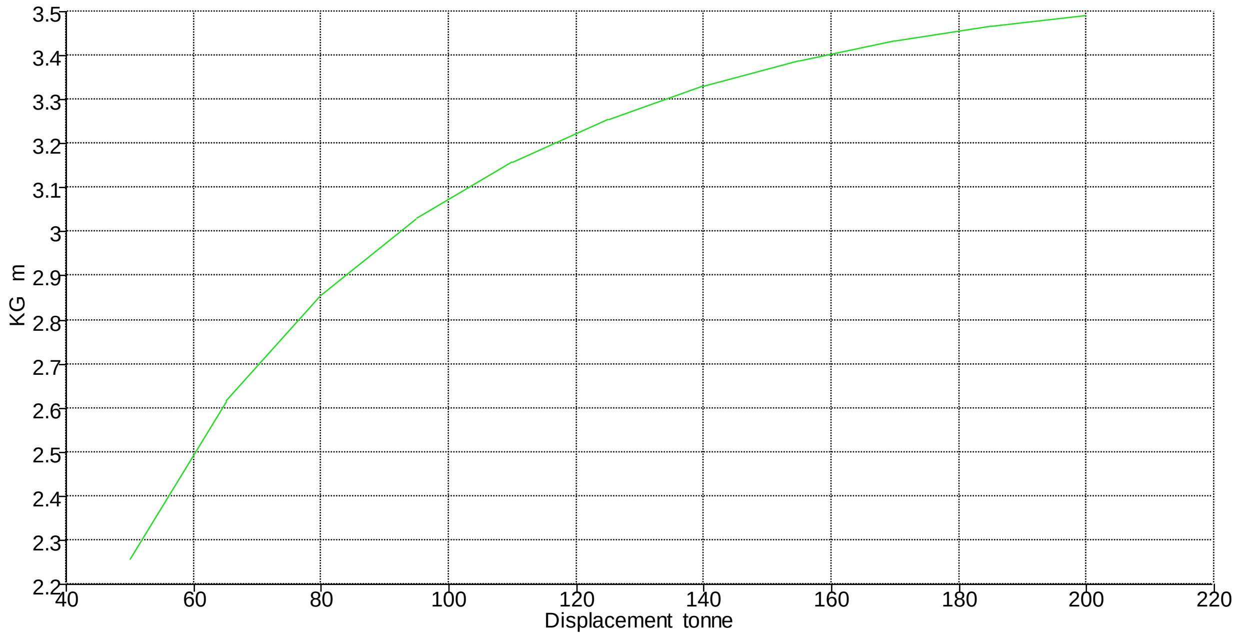
Fixed Trim = 0.5 m (+ve by stern)

Specific gravity = 1.025; (Density = 1.025 tonne/m^3)
Heel to starboard; heel range: from -50 deg to 90 deg in steps of 10 deg.

Criteria tested:

267(85) Ch2 - General Criteria	2.2.1: Area 0 to 30
267(85) Ch2 - General Criteria	2.2.1: Area 0 to 40
267(85) Ch2 - General Criteria	2.2.1: Area 30 to 40
267(85) Ch2 - General Criteria	2.2.2: Max GZ at 30 or greater
267(85) Ch2 - General Criteria	2.2.3: Angle of maximum GZ
267(85) Ch2 - General Criteria	2.2.4: Initial GMt
267(85) Ch2 - General Criteria	2.3: Severe wind and rolling

Displacement (intact) tonne	Draft Amidships m	LCG m	TCG m	VCG m	Limit KG m	min. GM m	Criterion	Name
50.00	0.987	17.909	0.000	2.259	2.259	3.904	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
65.00	1.084	17.030	0.000	2.616	2.616	4.480	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
80.00	1.173	16.431	0.000	2.857	2.857	3.725	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
95.00	1.259	16.046	0.000	3.030	3.030	3.087	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
110.0	1.343	15.786	0.000	3.158	3.158	2.566	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
125.0	1.424	15.606	0.000	3.255	3.255	2.147	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
140.0	1.504	15.478	0.000	3.330	3.330	1.807	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
155.0	1.583	15.387	0.000	3.388	3.388	1.529	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
170.0	1.661	15.322	0.000	3.432	3.432	1.300	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
185.0	1.737	15.276	0.000	3.465	3.465	1.110	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
200.0	1.813	15.245	0.000	3.492	3.492	0.948	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling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KG m
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220
Displacement tonne

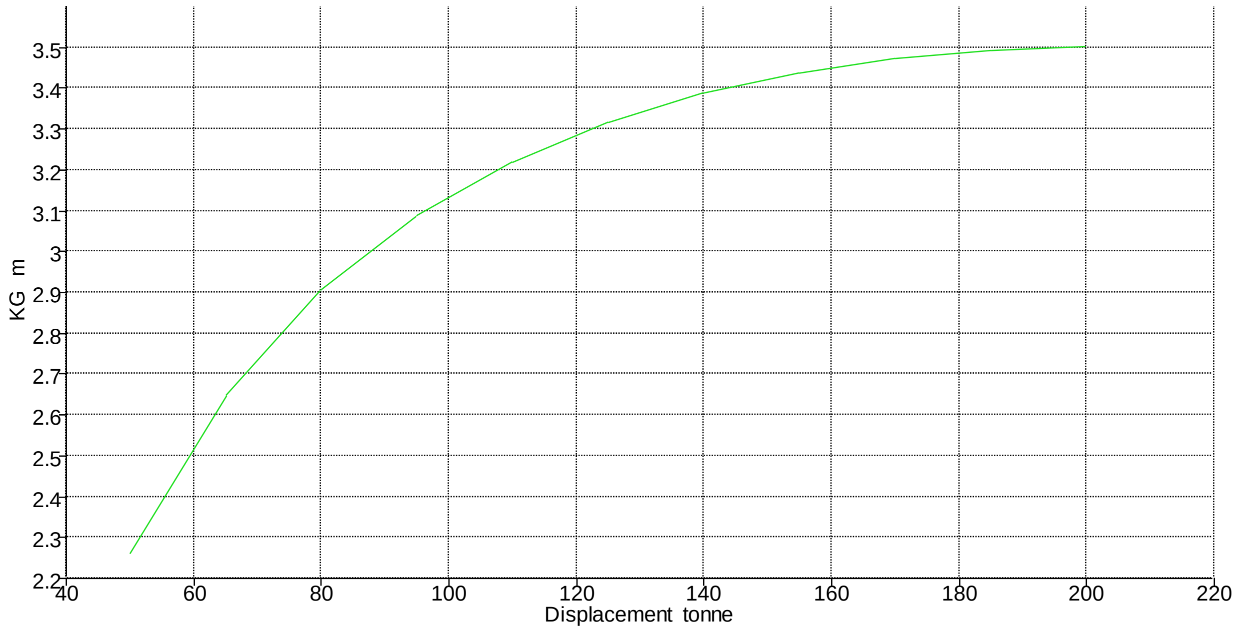
Fixed Trim = 1 m (+ve by stern)

Specific gravity = 1.025; (Density = 1.025 tonne/m^3)
Heel to starboard; heel range: from -50 deg to 90 deg in steps of 10 deg.

Criteria tested:

267(85) Ch2 - General Criteria	2.2.1: Area 0 to 30
267(85) Ch2 - General Criteria	2.2.1: Area 0 to 40
267(85) Ch2 - General Criteria	2.2.1: Area 30 to 40
267(85) Ch2 - General Criteria	2.2.2: Max GZ at 30 or greater
267(85) Ch2 - General Criteria	2.2.3: Angle of maximum GZ
267(85) Ch2 - General Criteria	2.2.4: Initial GMt
267(85) Ch2 - General Criteria	2.3: Severe wind and rolling

Displacement (intact) tonne	Draft Amidships m	LCG m	TCG m	VCG m	Limit KG m	min. GM m	Criterion	Name
50.00	0.924	14.918	0.000	2.262	2.262	5.711	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
65.00	1.021	14.477	0.000	2.648	2.648	4.524	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
80.00	1.113	14.250	0.000	2.908	2.908	3.666	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
95.00	1.201	14.131	0.000	3.089	3.089	3.001	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
110.0	1.287	14.073	0.000	3.220	3.220	2.469	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
125.0	1.371	14.050	0.000	3.317	3.317	2.047	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
140.0	1.453	14.050	0.000	3.388	3.388	1.710	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
155.0	1.534	14.064	0.000	3.438	3.438	1.439	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
170.0	1.613	14.088	0.000	3.473	3.473	1.219	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
185.0	1.691	14.118	0.000	3.493	3.493	1.042	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling
200.0	1.768	14.153	0.000	3.503	3.503	0.897	267(85) Ch2 - General Criteria	2.3: Severe wind and rolling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KG m
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Displacement tonne